```
Supersonic validated modal points — 50 points

Spectral CSV : assets/classic_supersonic/final_50pts/supersonic_modal_spectral_validated_50pts.csv
Fields CSV   : assets/classic_supersonic/final_50pts/supersonic_modal_fields_p_rho_u_v_50pts.csv
Overlay PNG  : supersonic_50pts_overlay_blumen_vs_validated.png

Nombre total de points : 50

Répartition par Mach :
  - M=1.10 : 2 points
  - M=1.20 : 6 points
  - M=1.25 : 1 points
  - M=1.30 : 9 points
  - M=1.33 : 1 points
  - M=1.40 : 11 points
  - M=1.50 : 8 points
  - M=1.60 : 2 points
  - M=1.70 : 4 points
  - M=1.80 : 3 points
  - M=1.90 : 3 points

Audit champs modaux :
  - base_fields_match_spectral : 28
  - base_fields_spectral_mismatch_requires_review : 16
  - extension_exact_shooting_fields : 6
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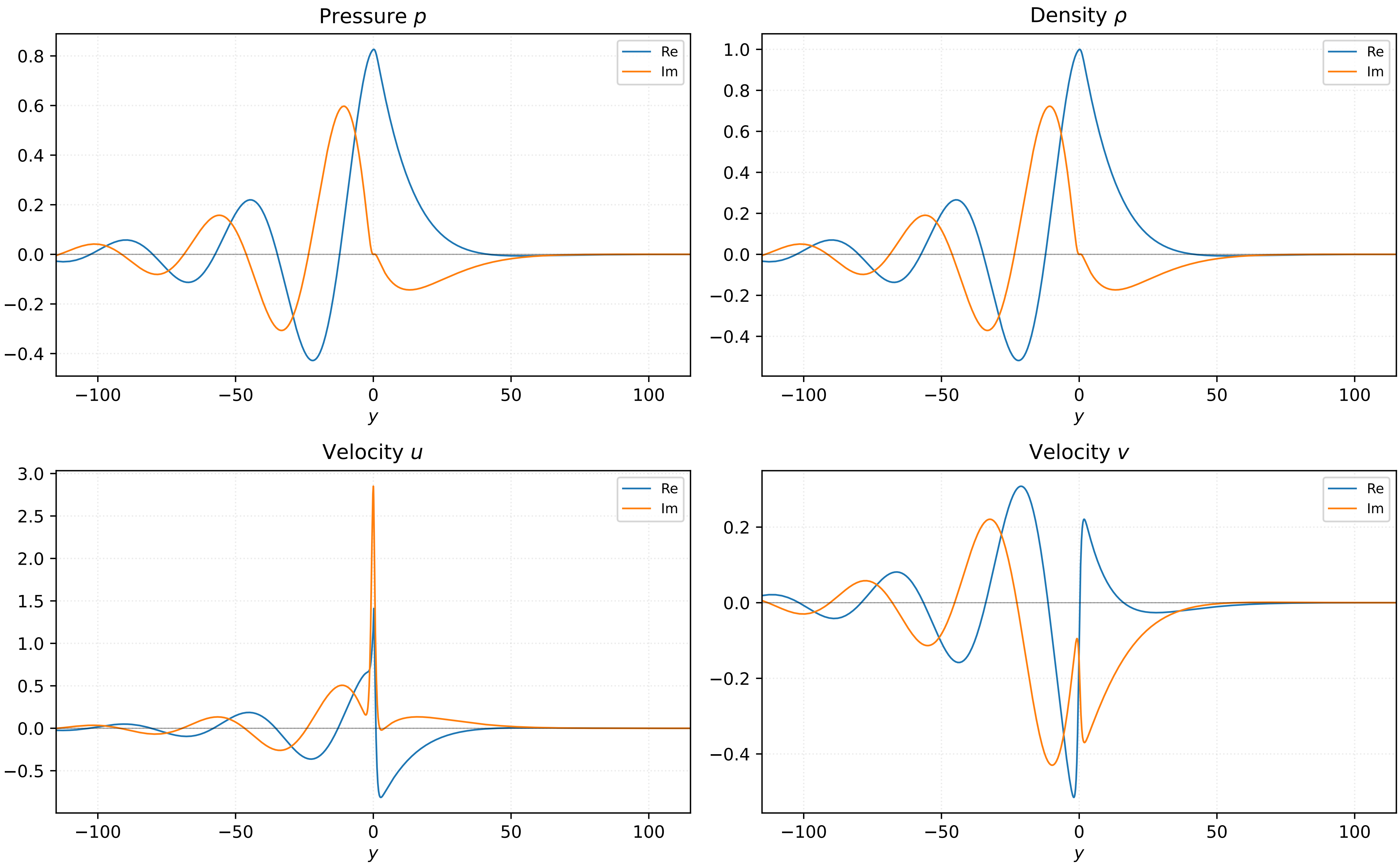
M=1.10, alpha=0.175000, c_r=0.150814, c_i=0.092213, omega_i=1.613732e-02



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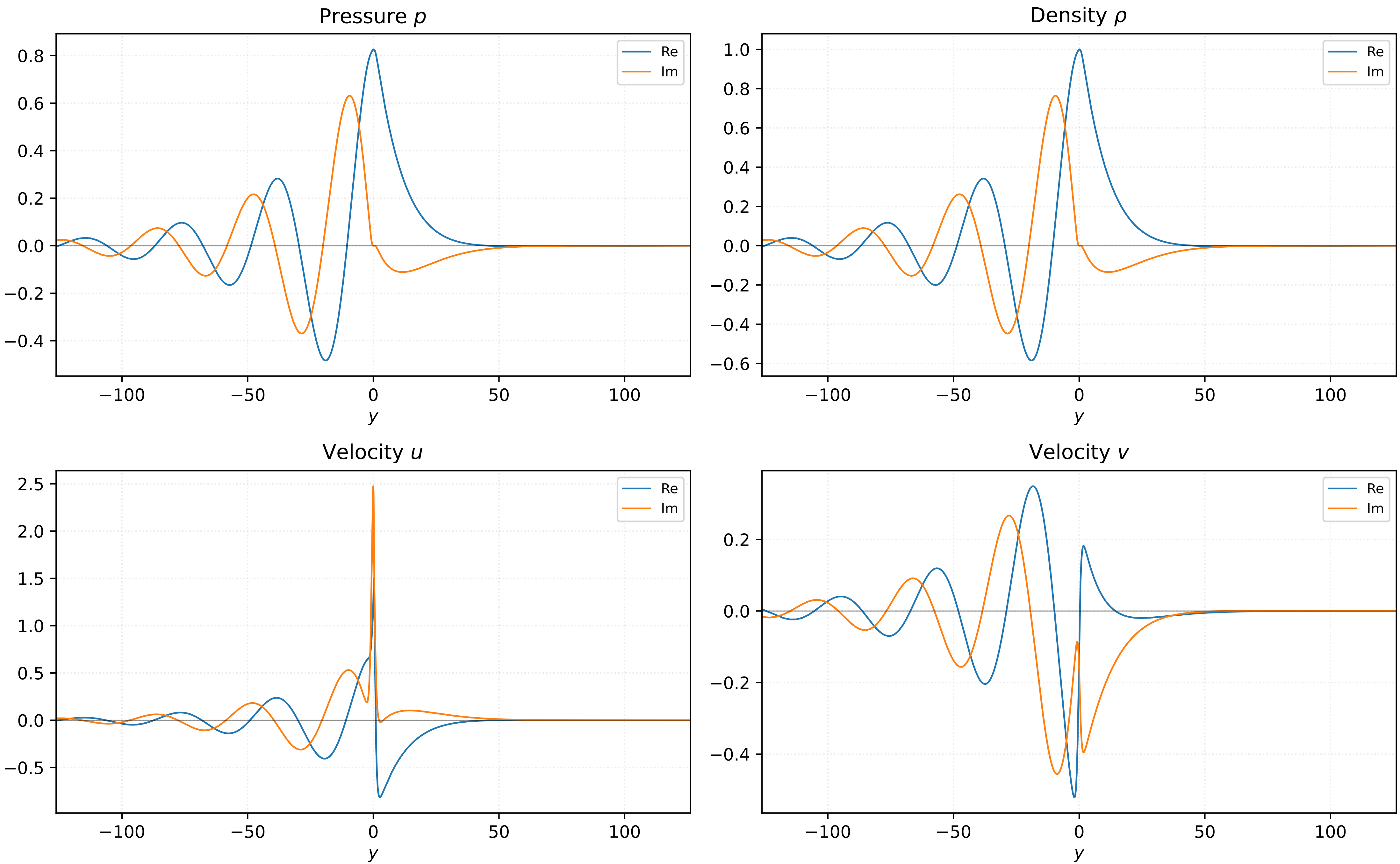
M=1.10, alpha=0.200000, c_r=0.165676, c_i=0.079180, omega_i=1.583593e-02



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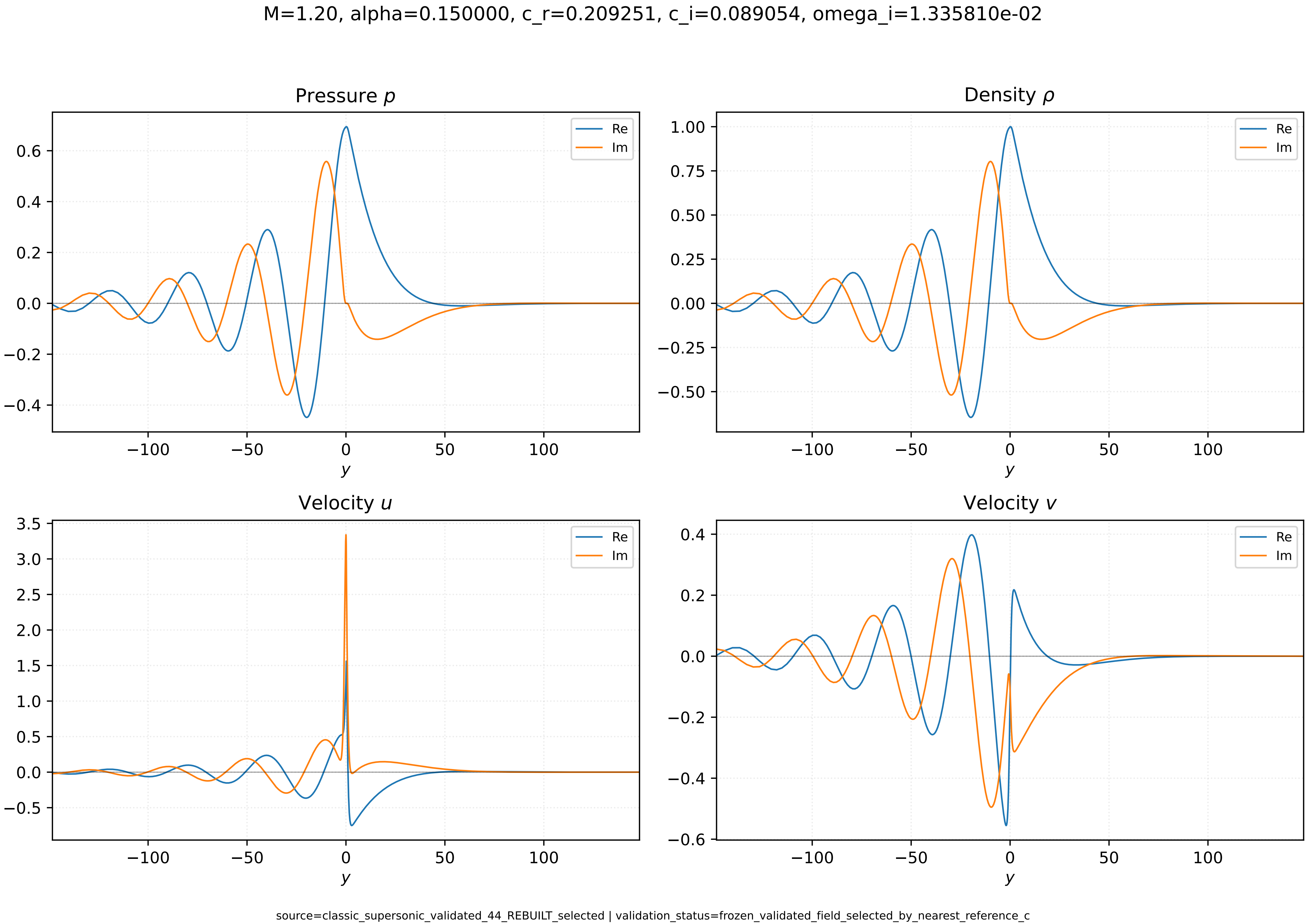
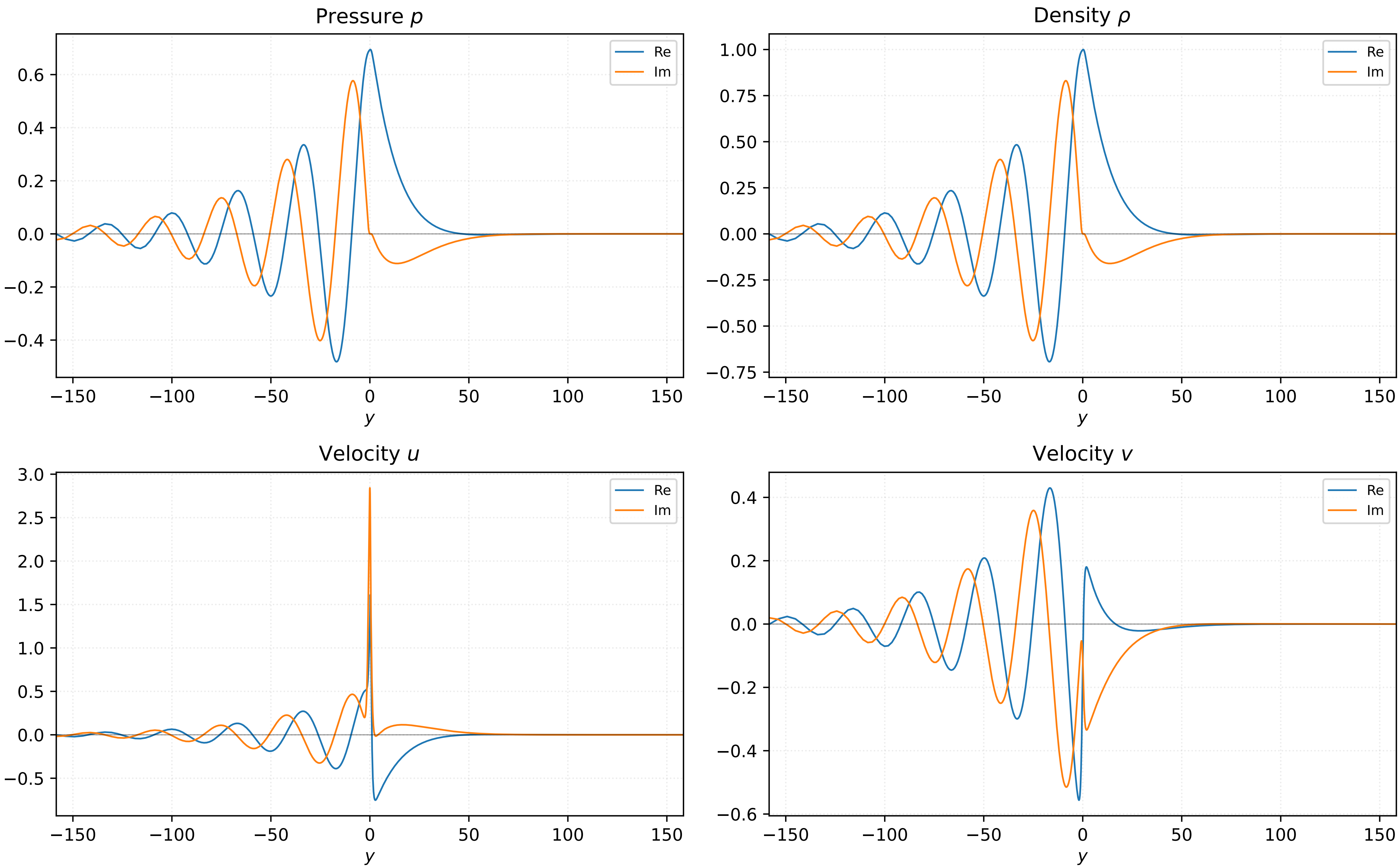


M=1.20, alpha=0.150000, c_r=0.209251, c_i=0.089054, omega_i=1.335810e-02
Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
0.6
0.4
0.2
0.0
−0.2
−0.4
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0.4
0.2
0.0
−0.2
−0.4
−0.6
−100
−50
0
50
100
y
source=classic_supersonic_validated_44_REBUILT_selected | validation_status=frozen_validated_field_selected_by_nearest_reference_c

M=1.20, alpha=0.175000, c_r=0.225911, c_i=0.076243, omega_i=1.334250e-02



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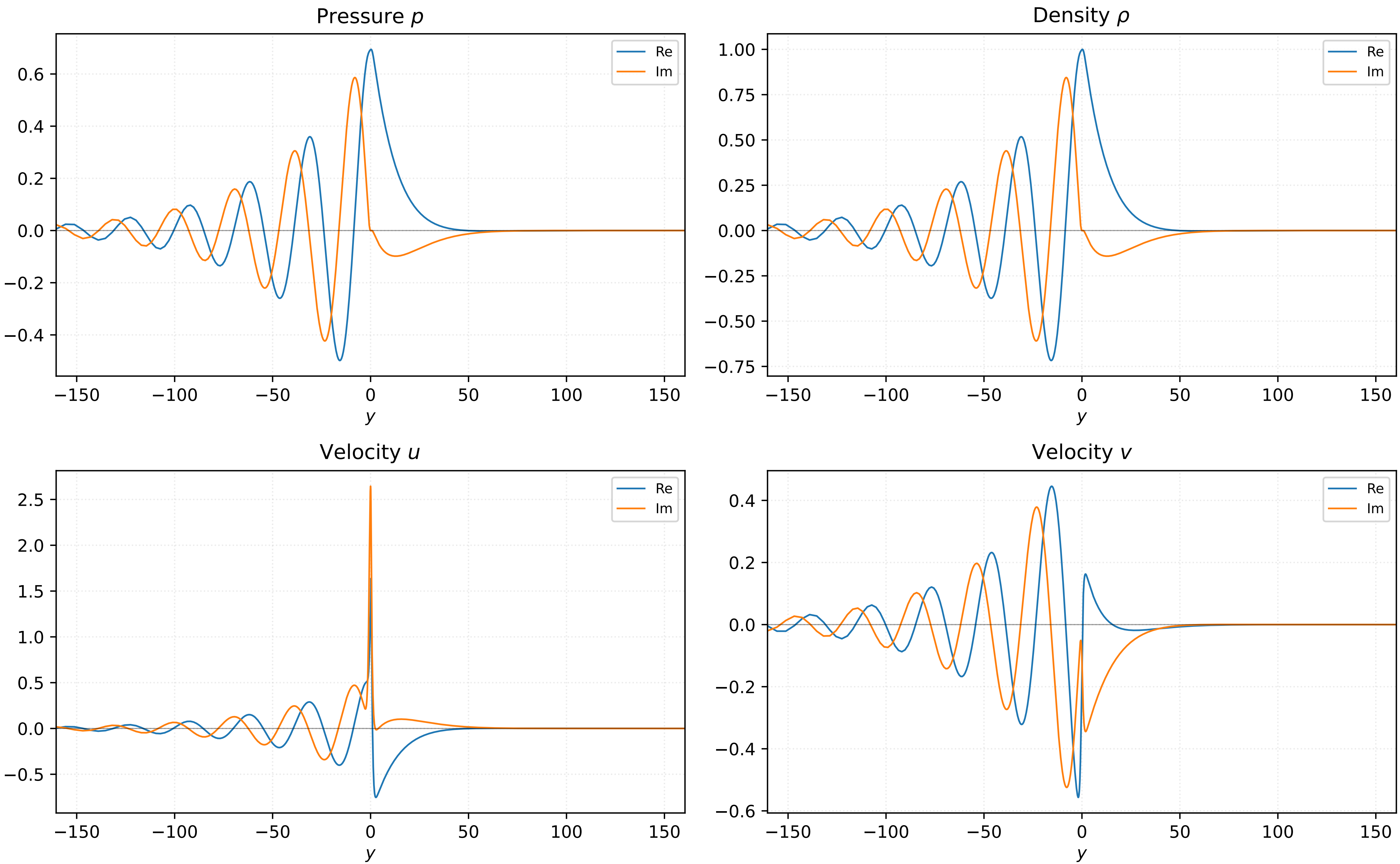

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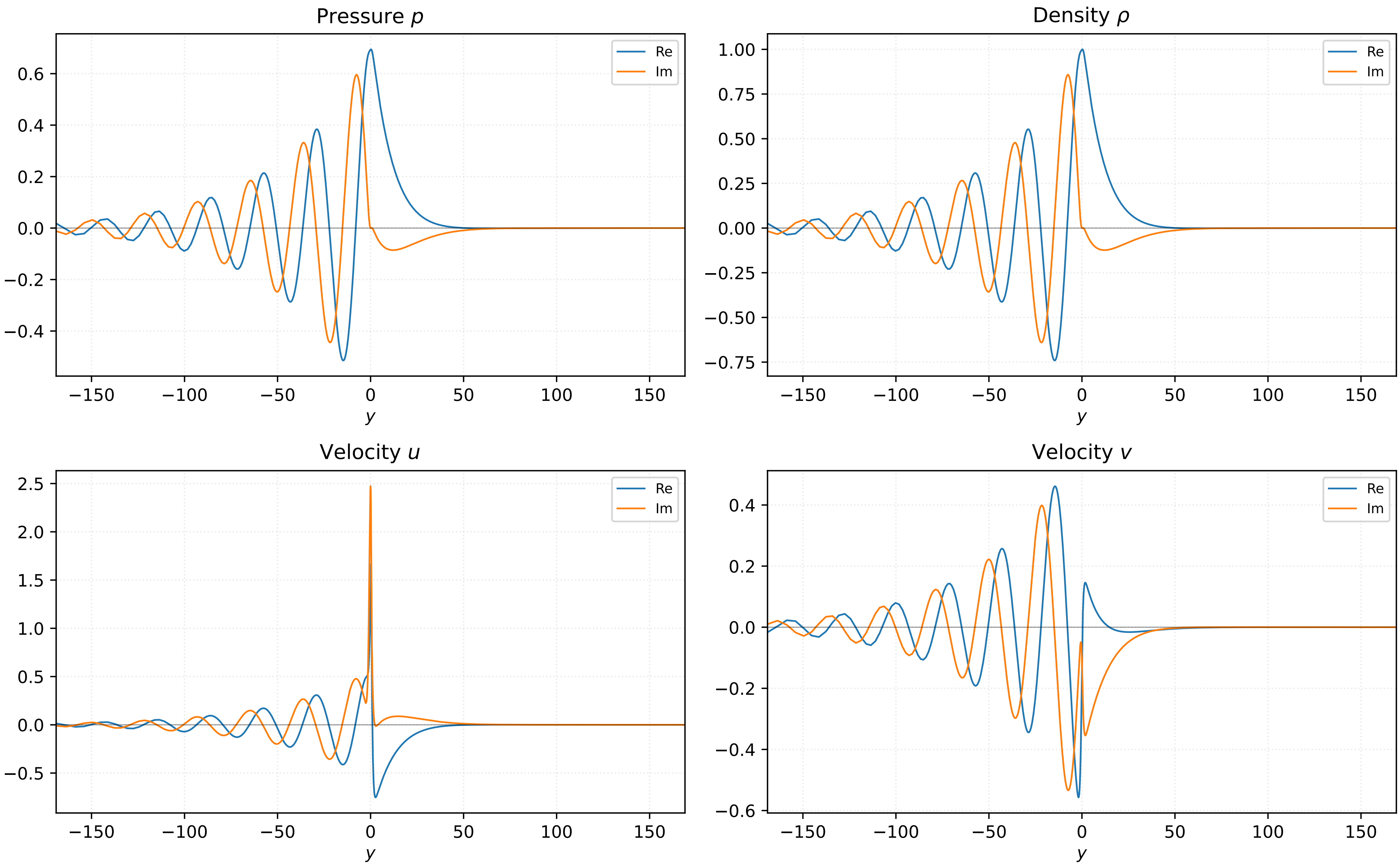

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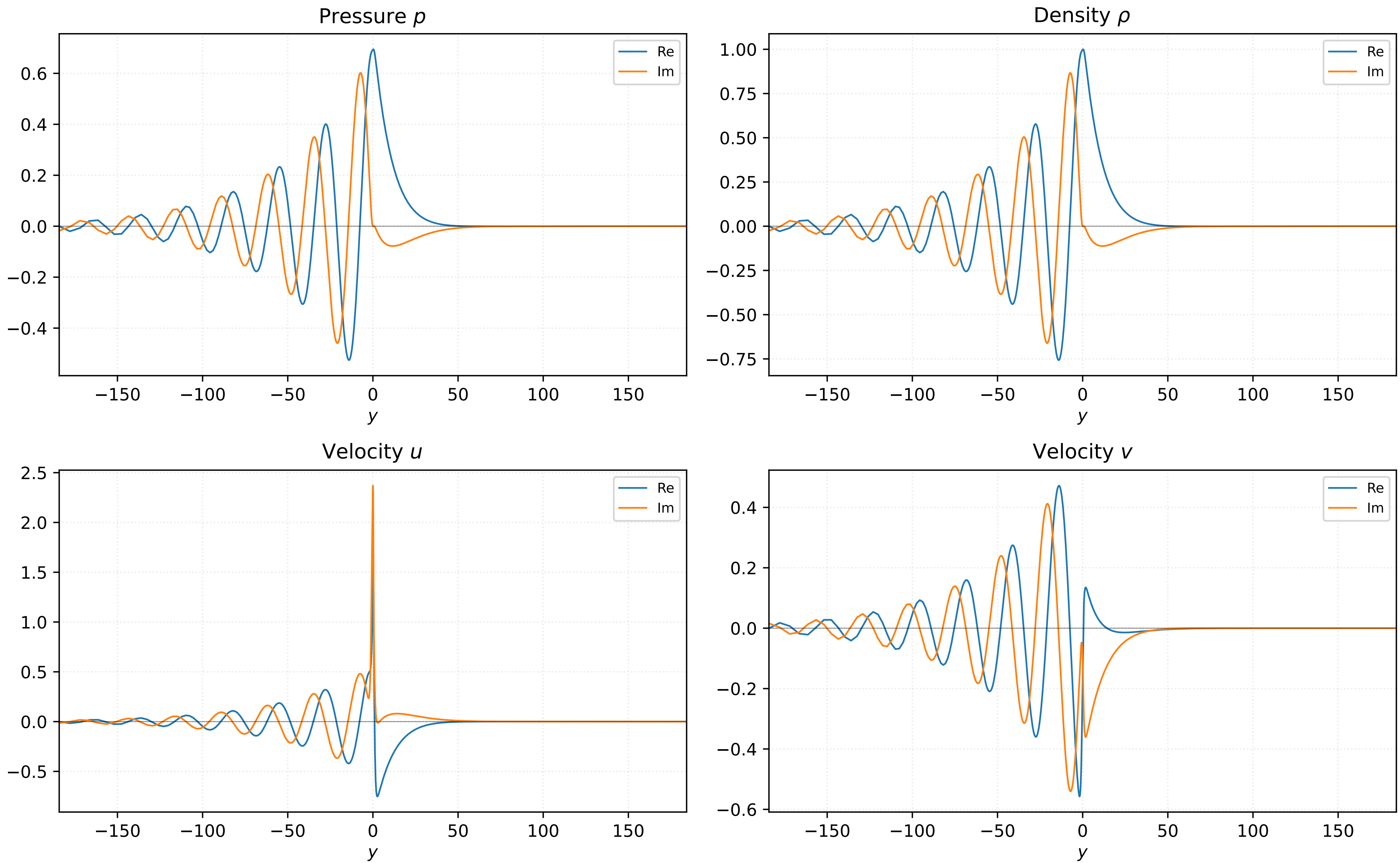

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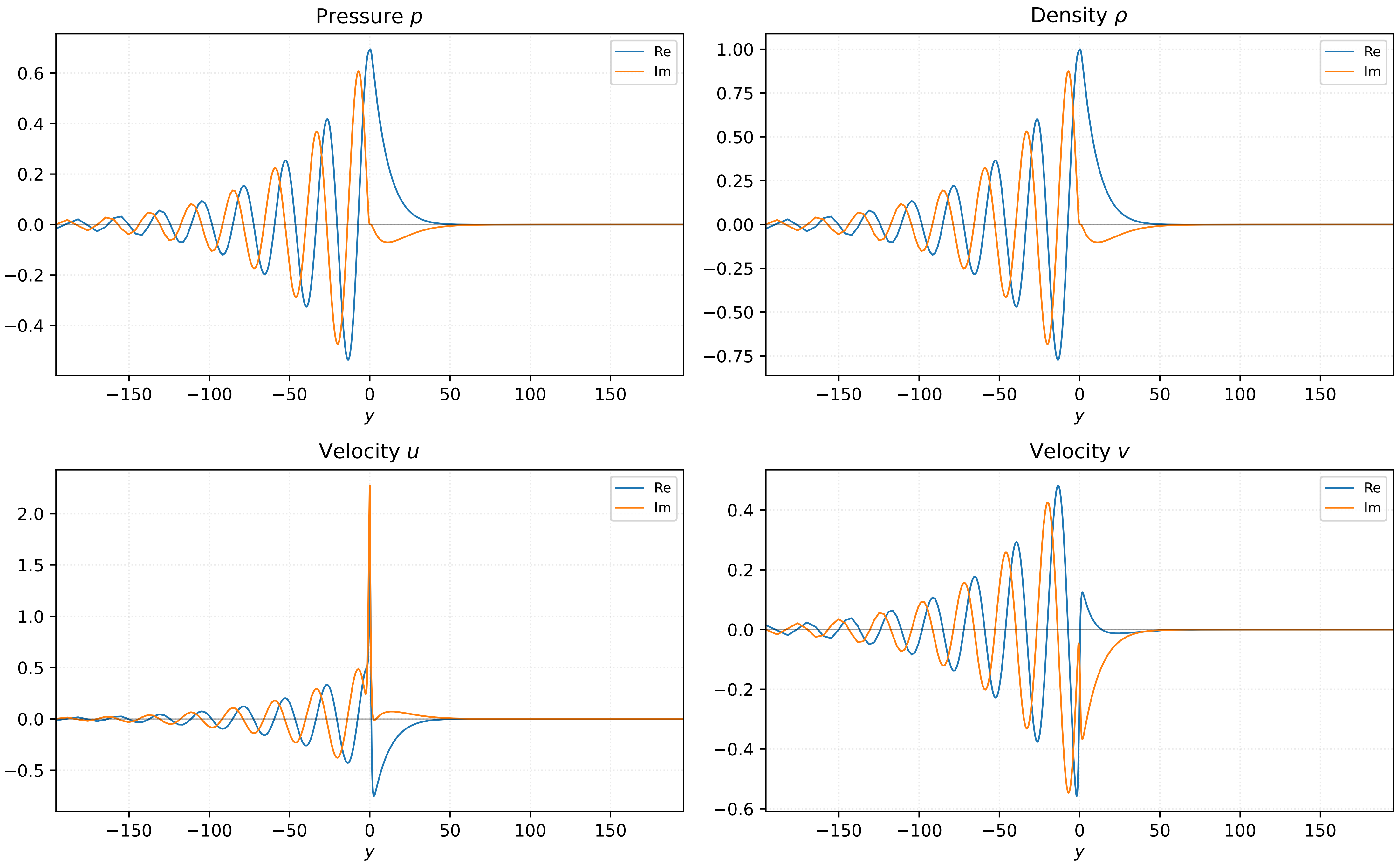

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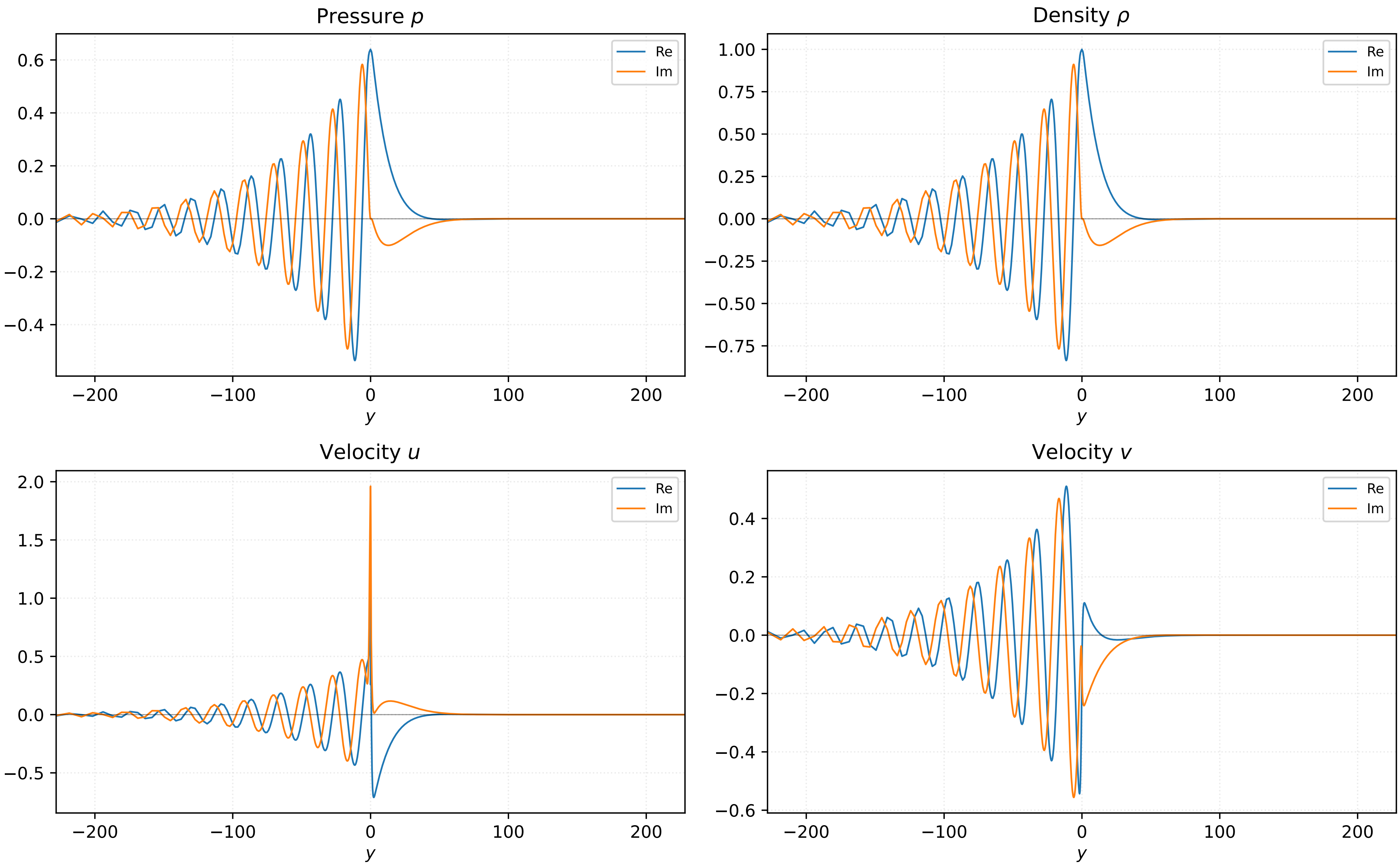

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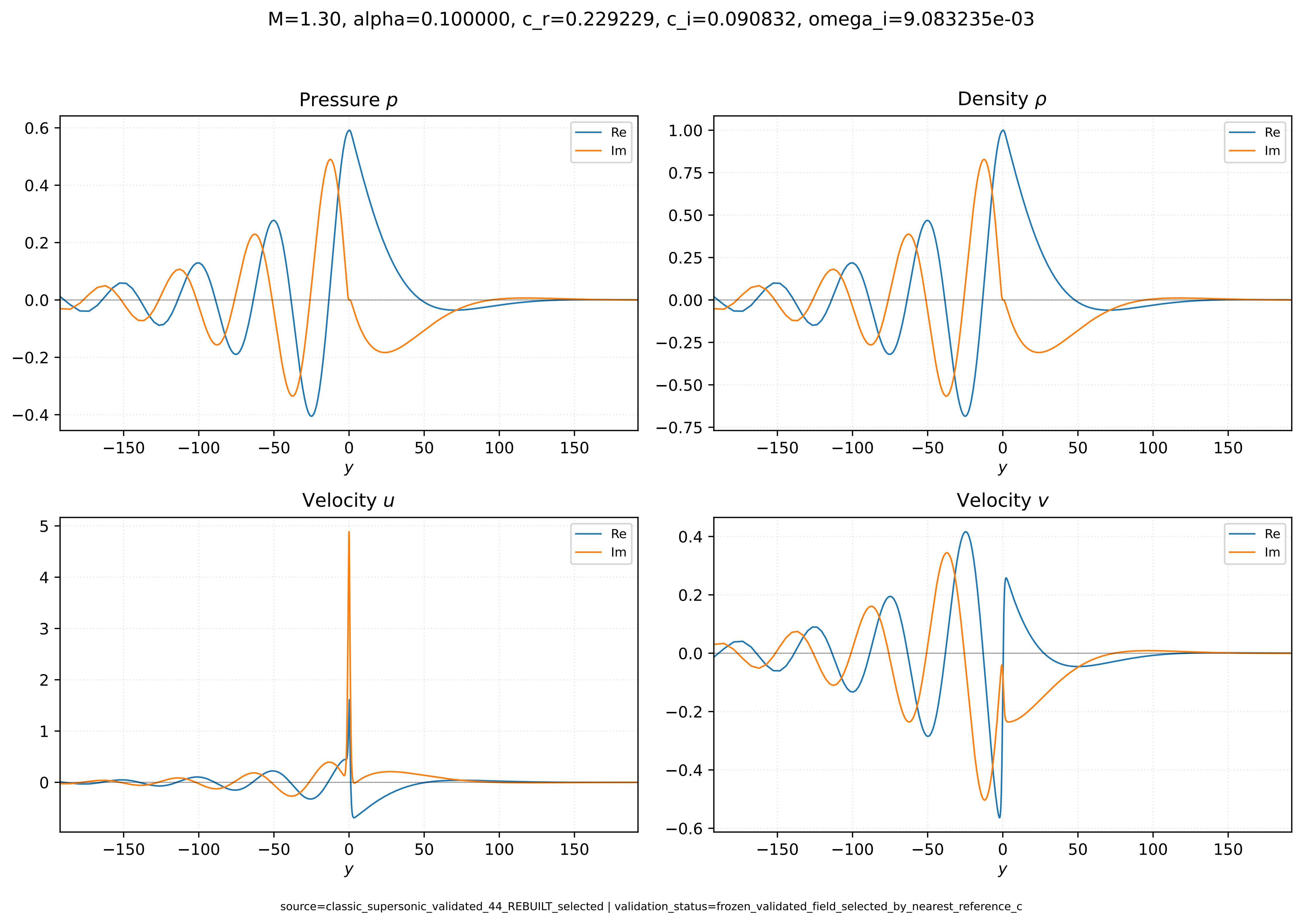
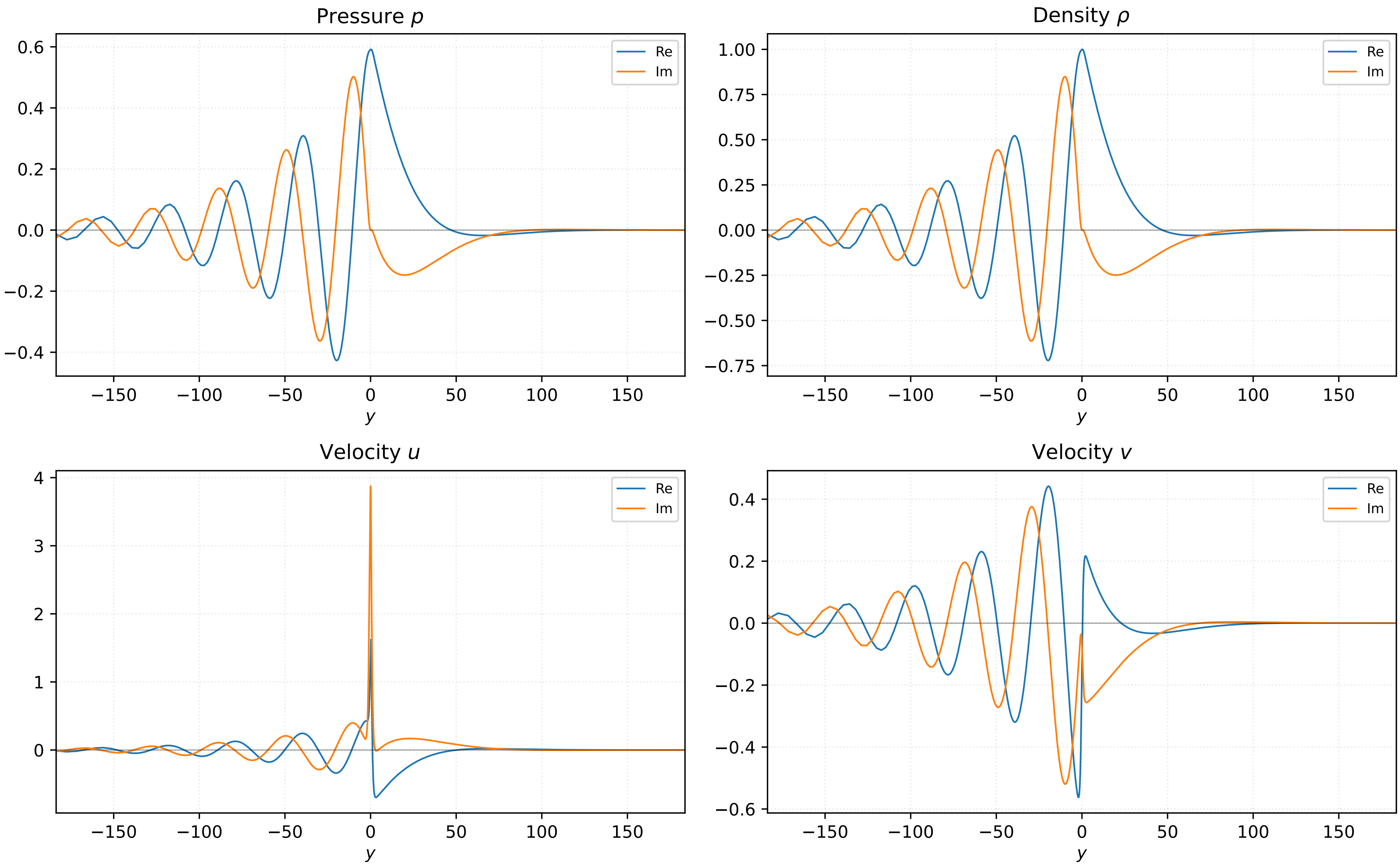

M=1.30, alpha=0.100000, c_r=0.229229, c_i=0.090832, omega_i=9.083235e-03
Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
y
source=classic_supersonic_validated_44_REBUILT_selected | validation_status=frozen_validated_field_selected_by_nearest_reference_c

M=1.30, alpha=0.125000, c_r=0.250732, c_i=0.080562, omega_i=1.007028e-02



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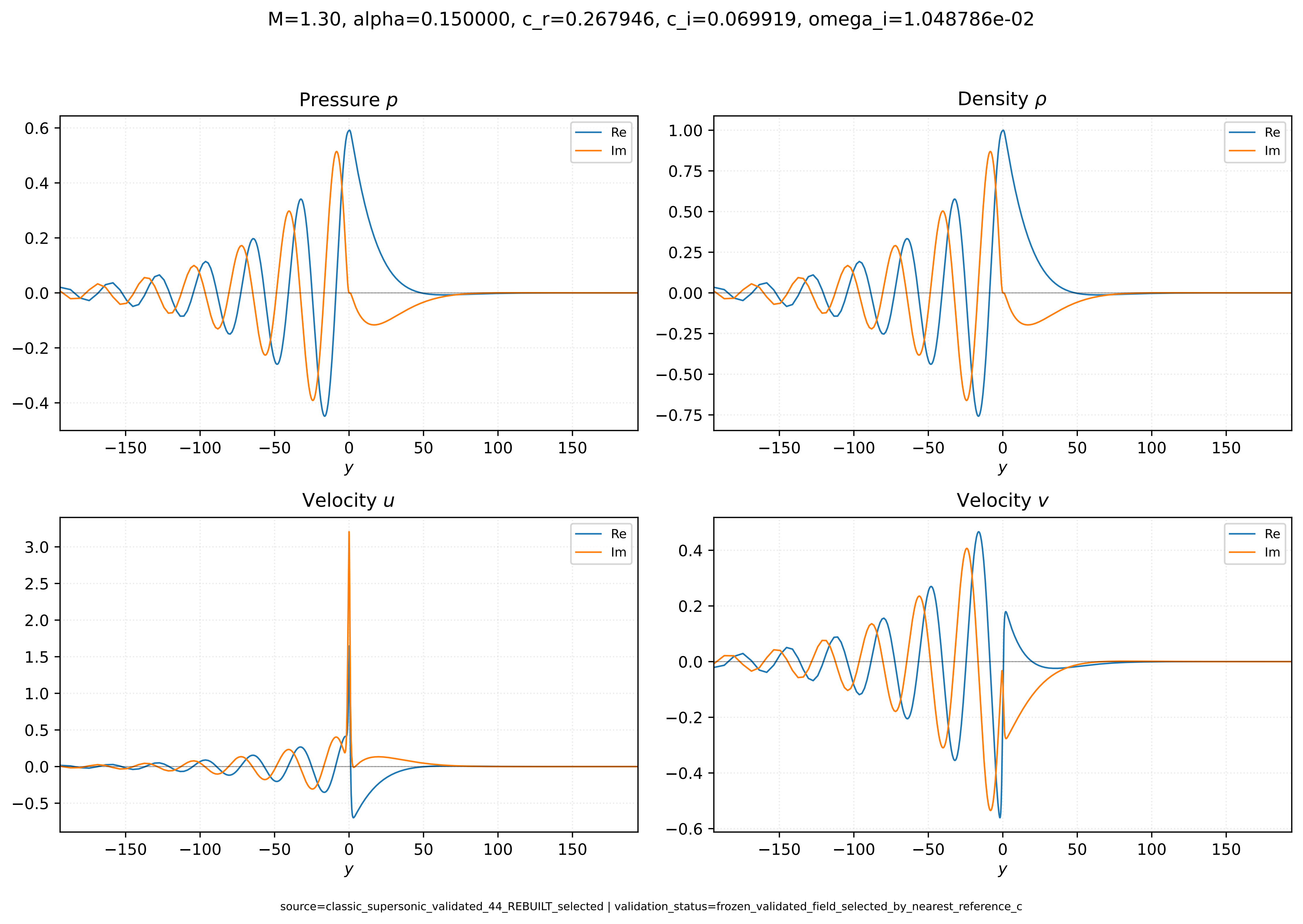

M=1.30, alpha=0.150000, c_r=0.267946, c_i=0.069919, omega_i=1.048786e-02
Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
y
source=classic_supersonic_validated_44_REBUILT_selected | validation_status=frozen_validated_field_selected_by_nearest_reference_c

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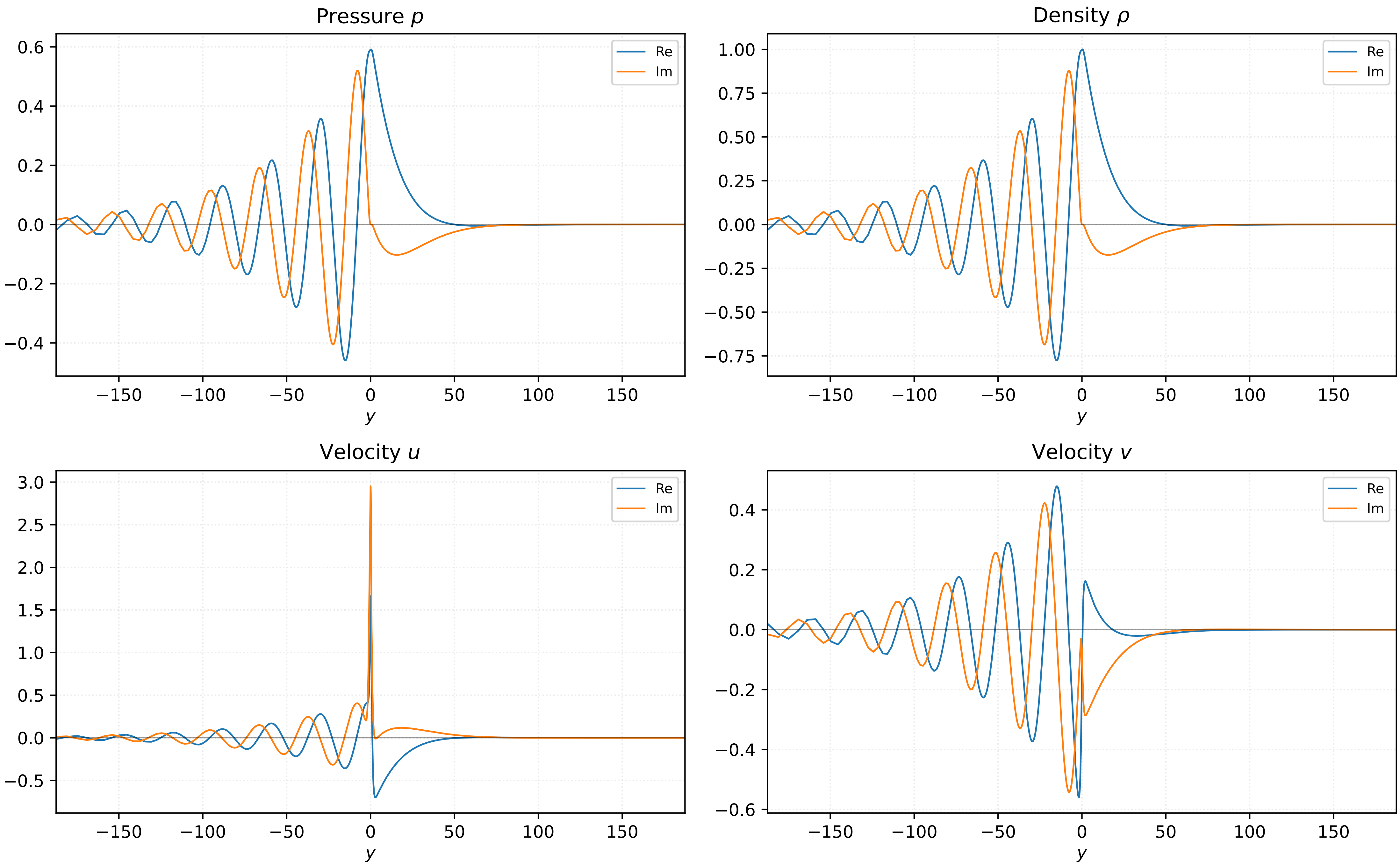

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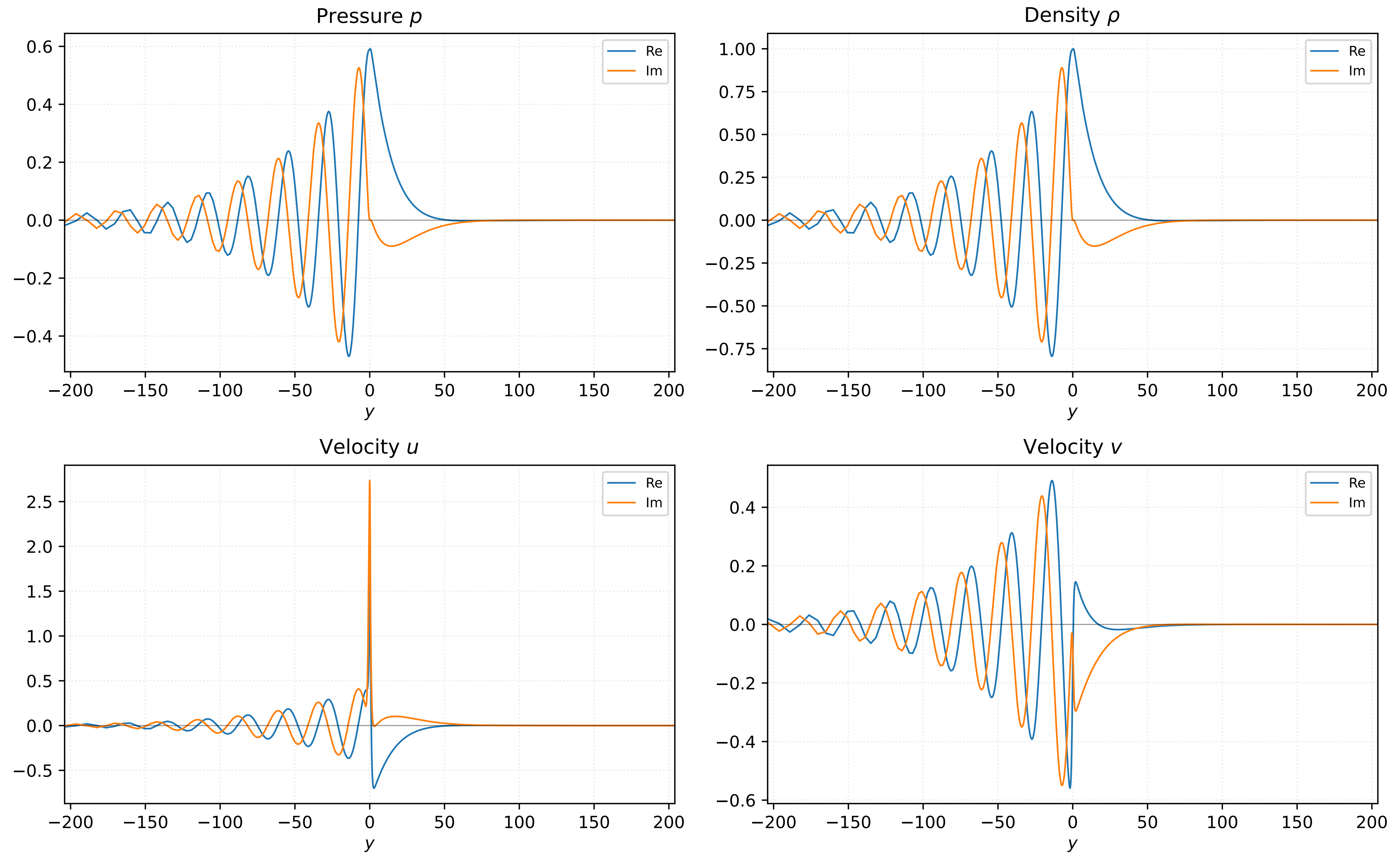

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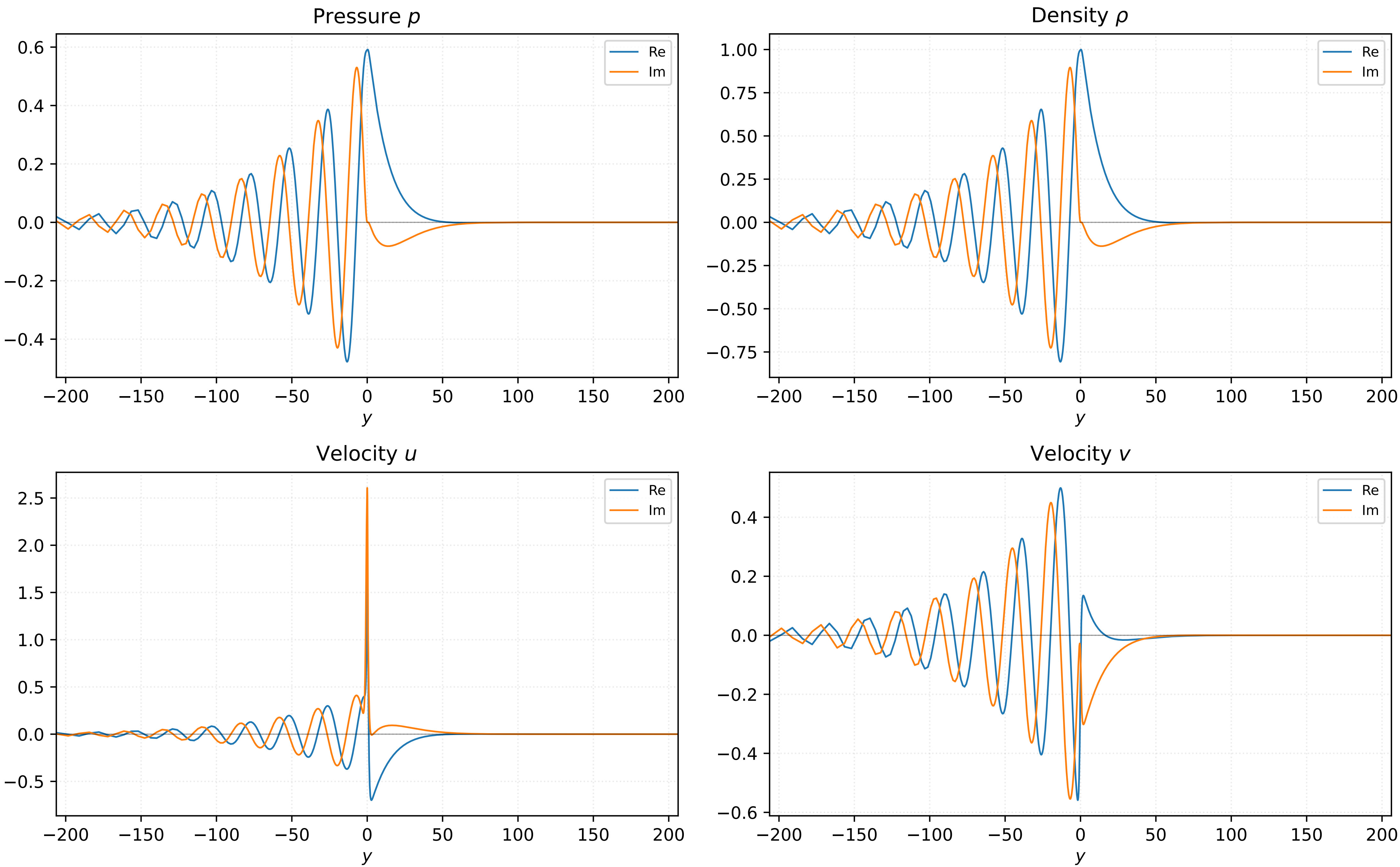

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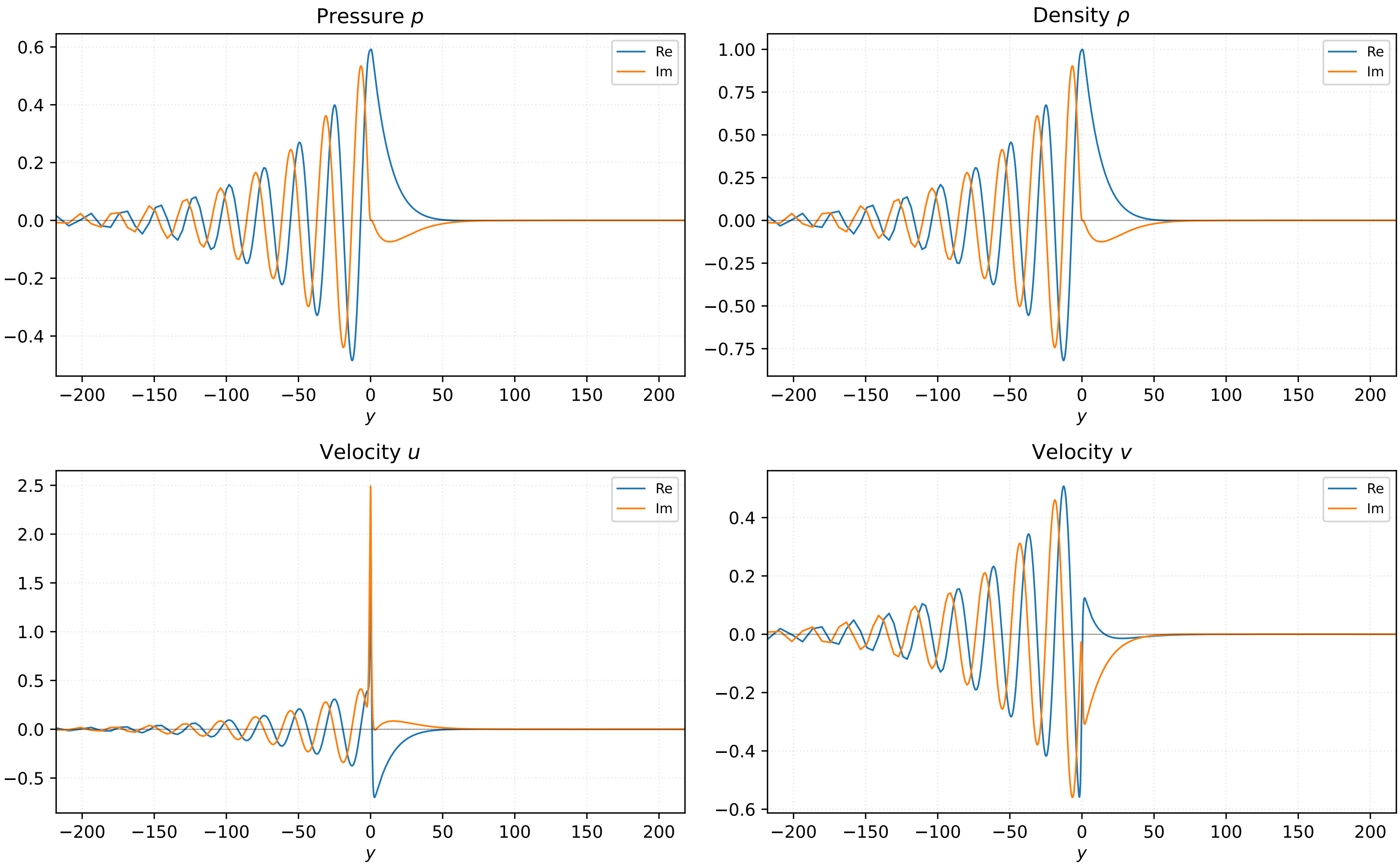

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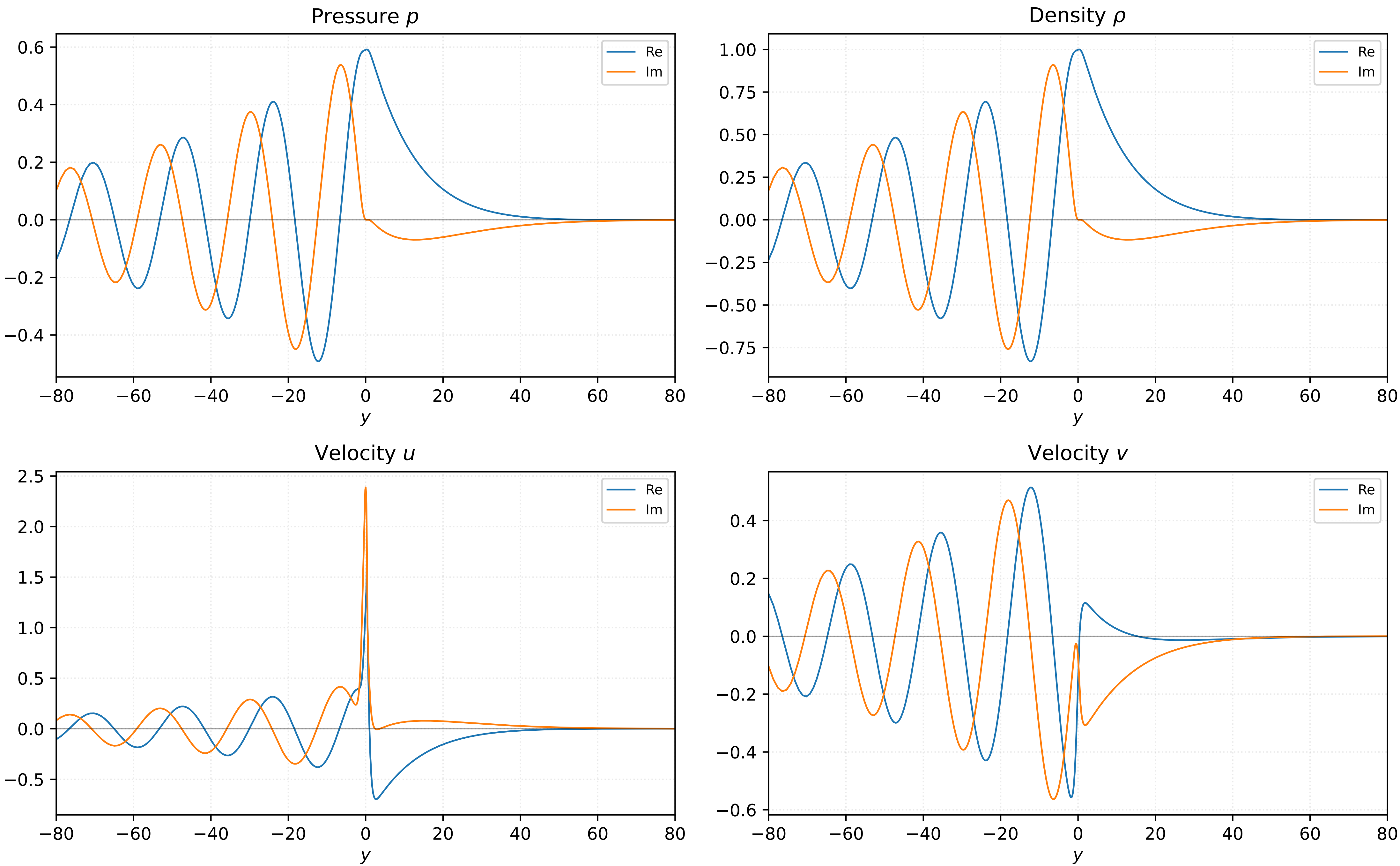

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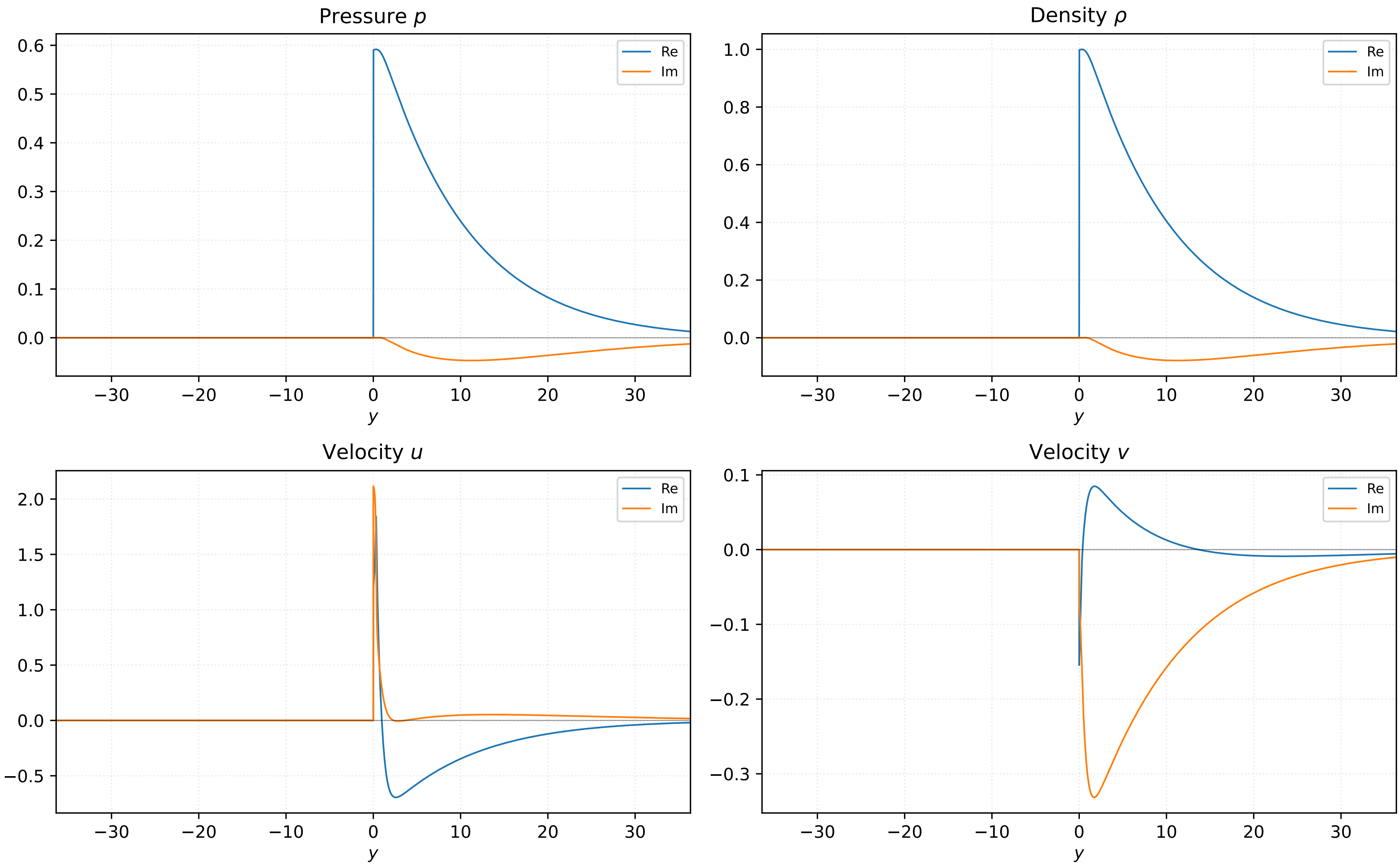

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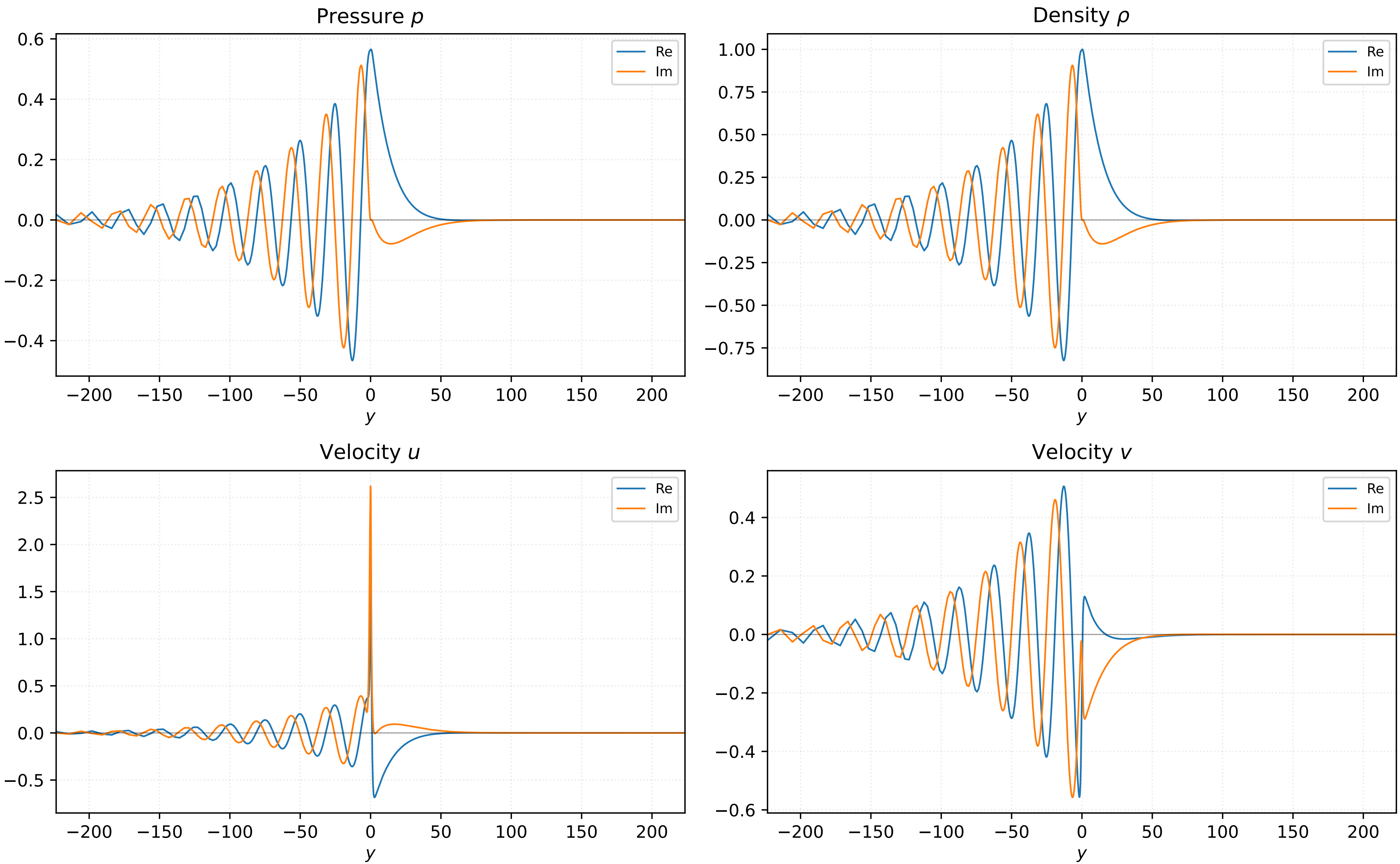

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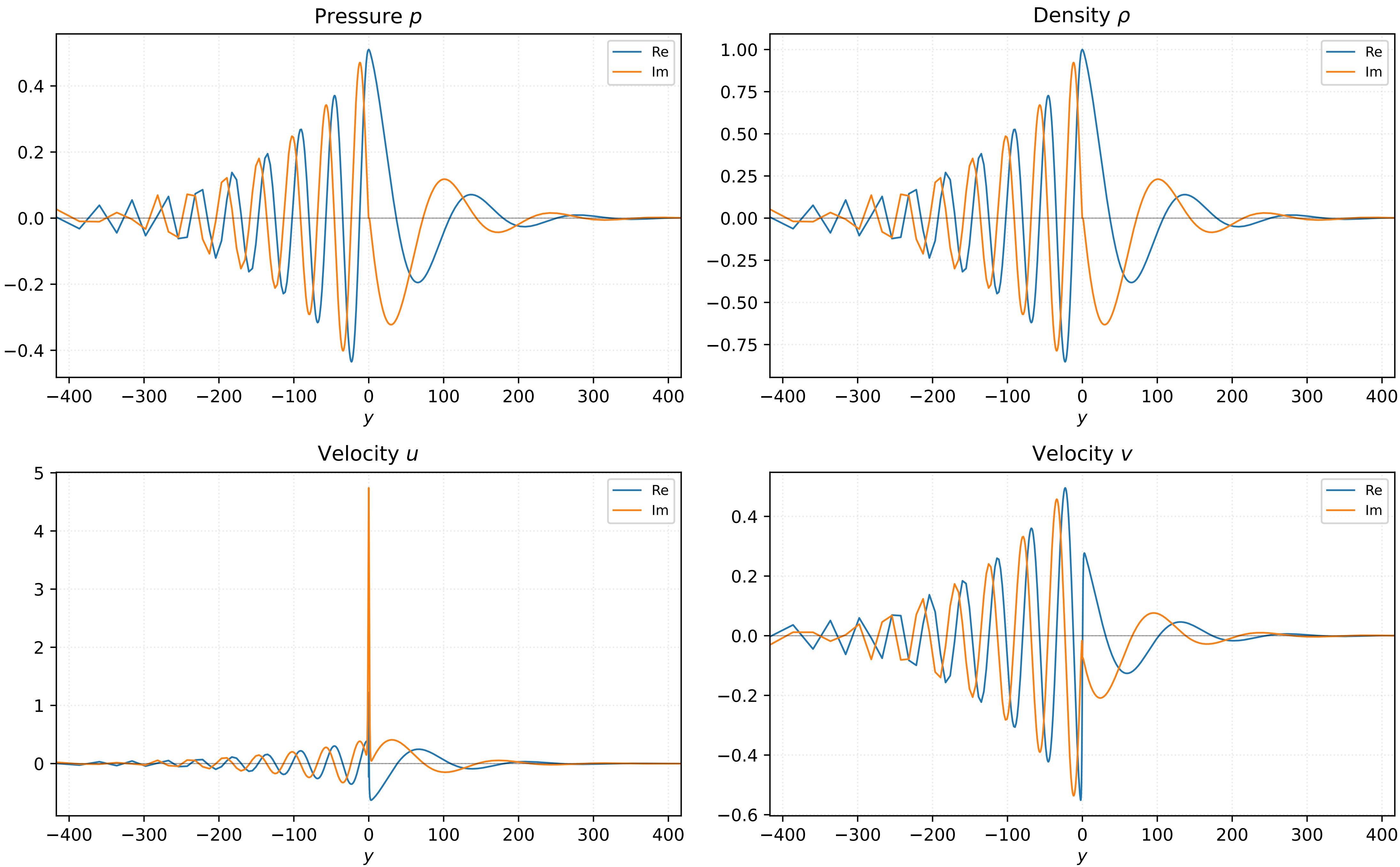

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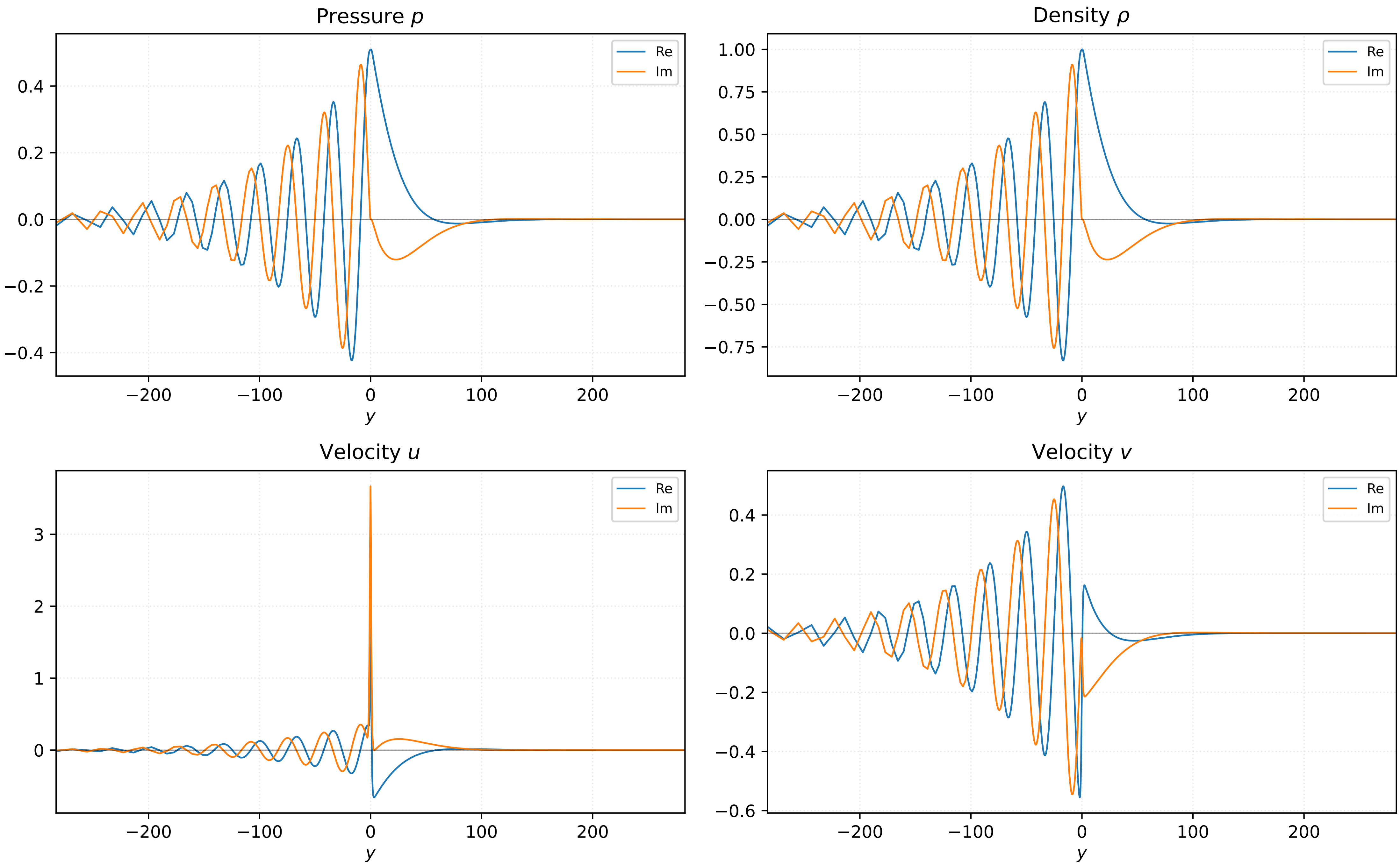

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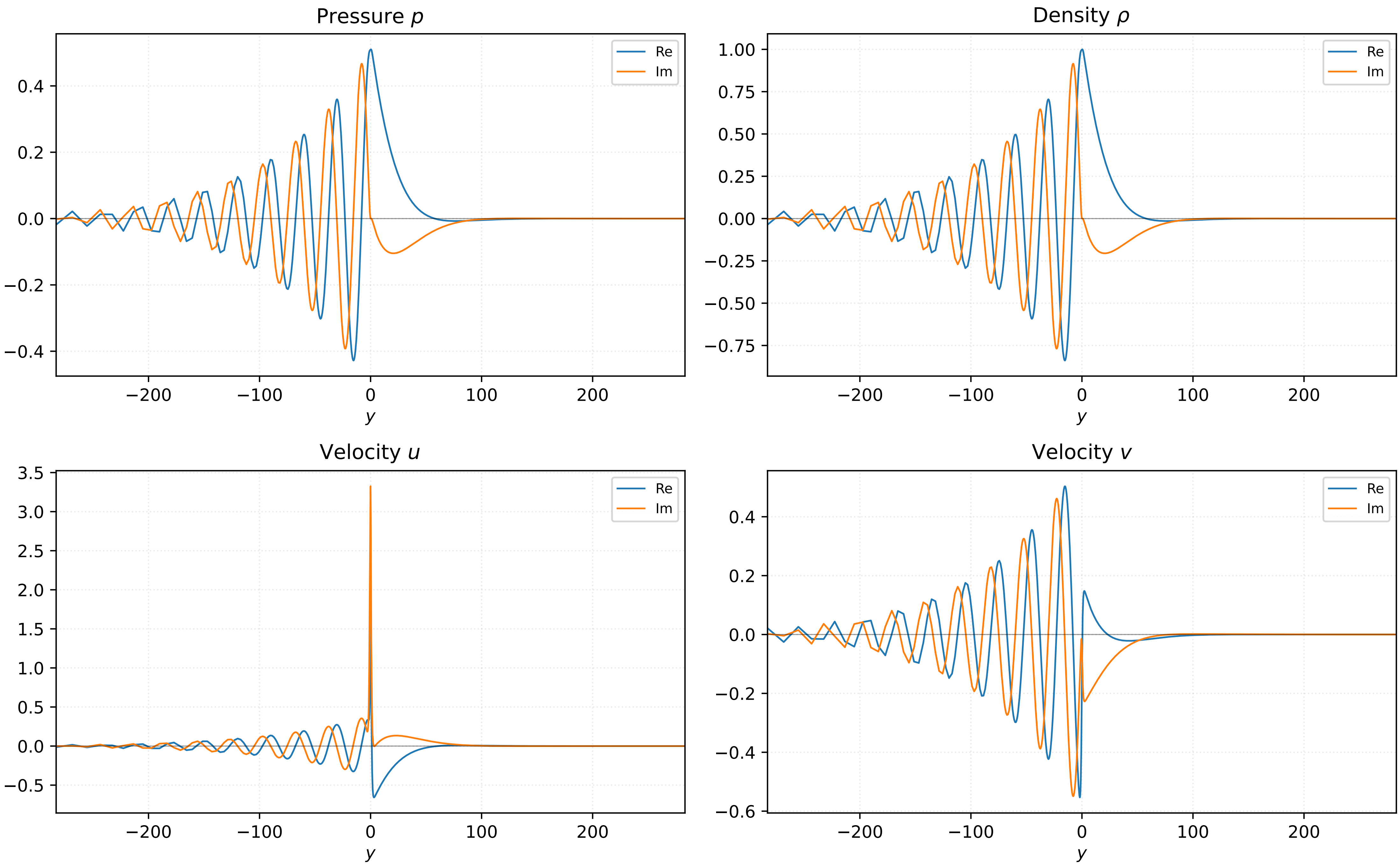

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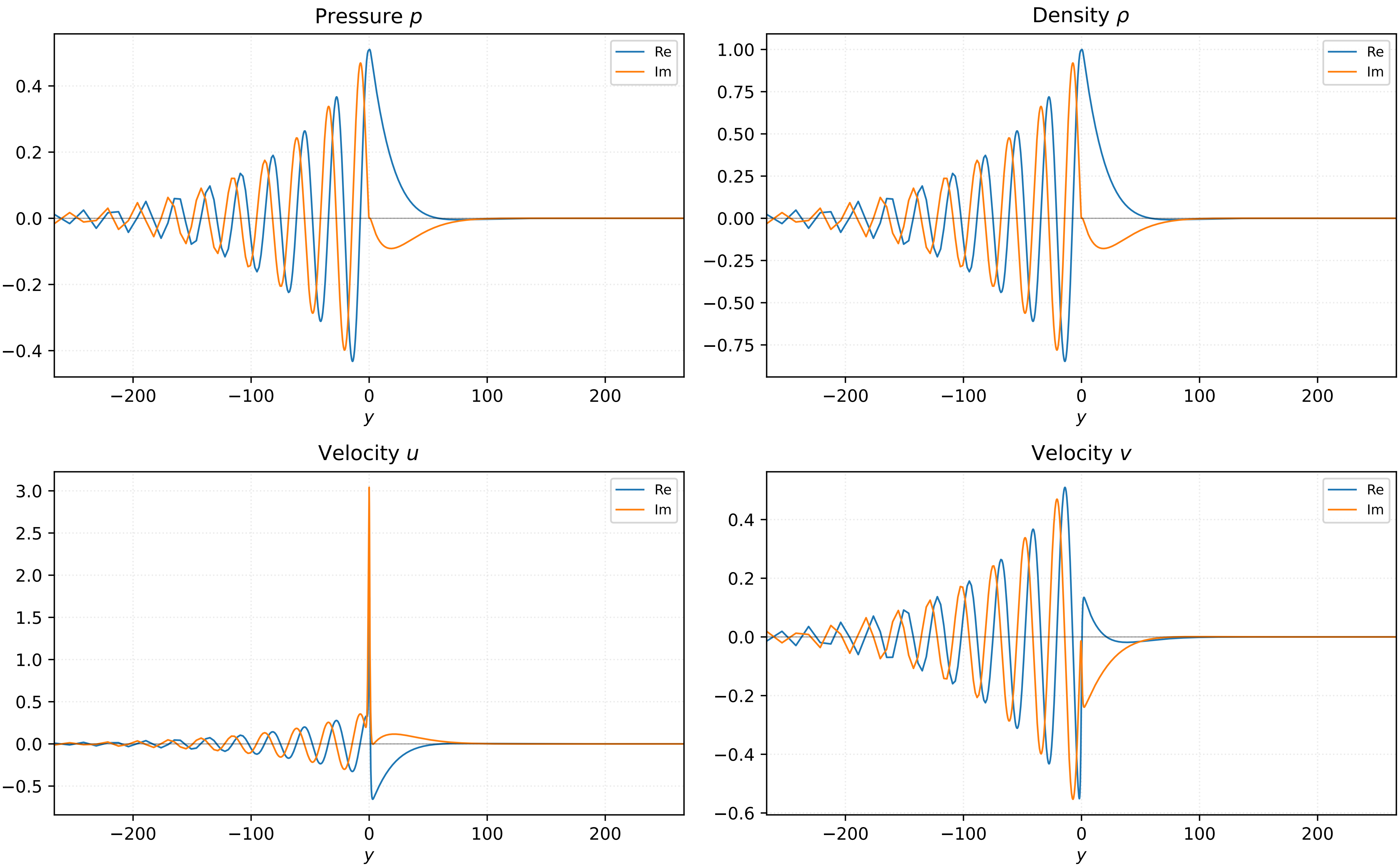

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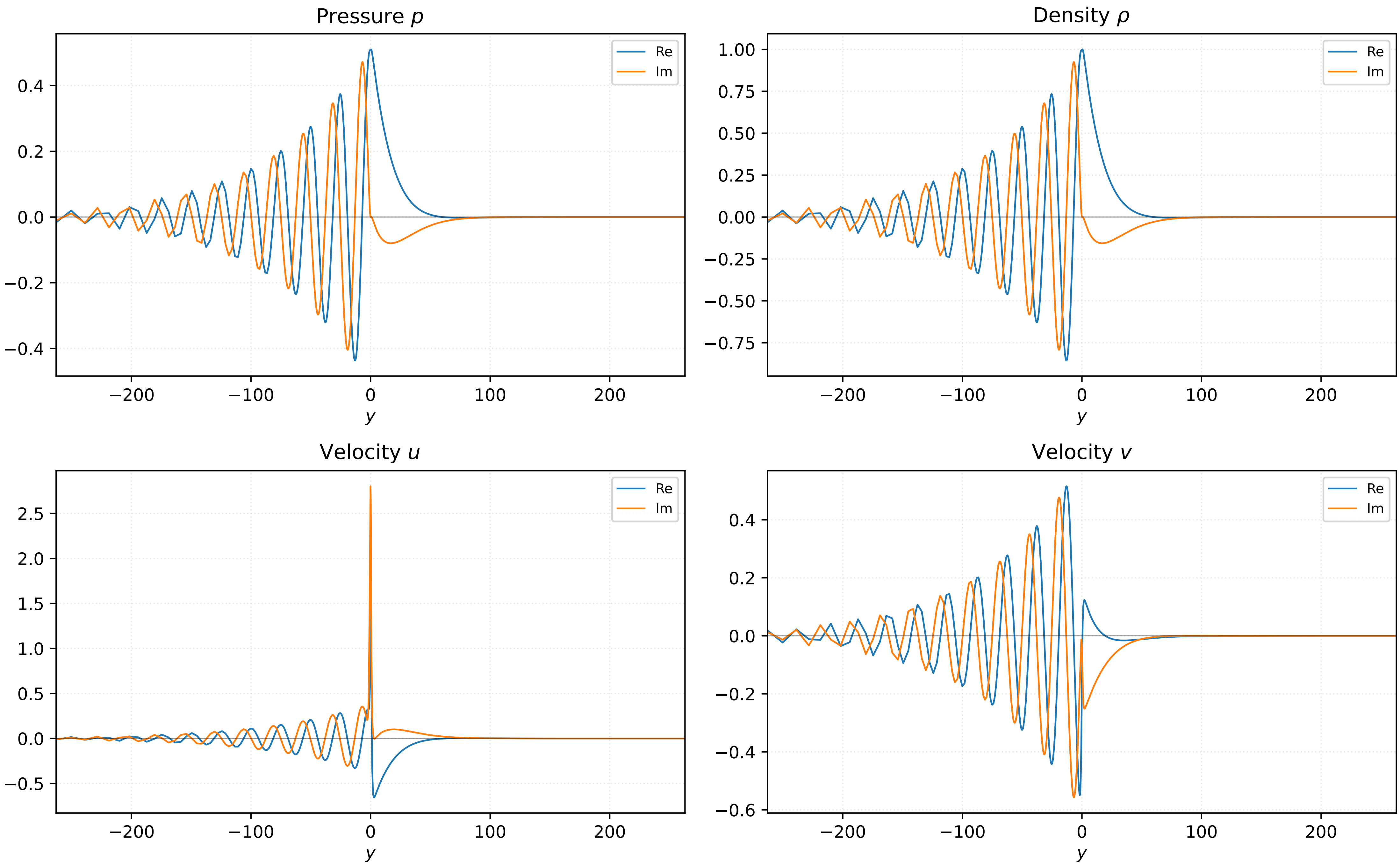

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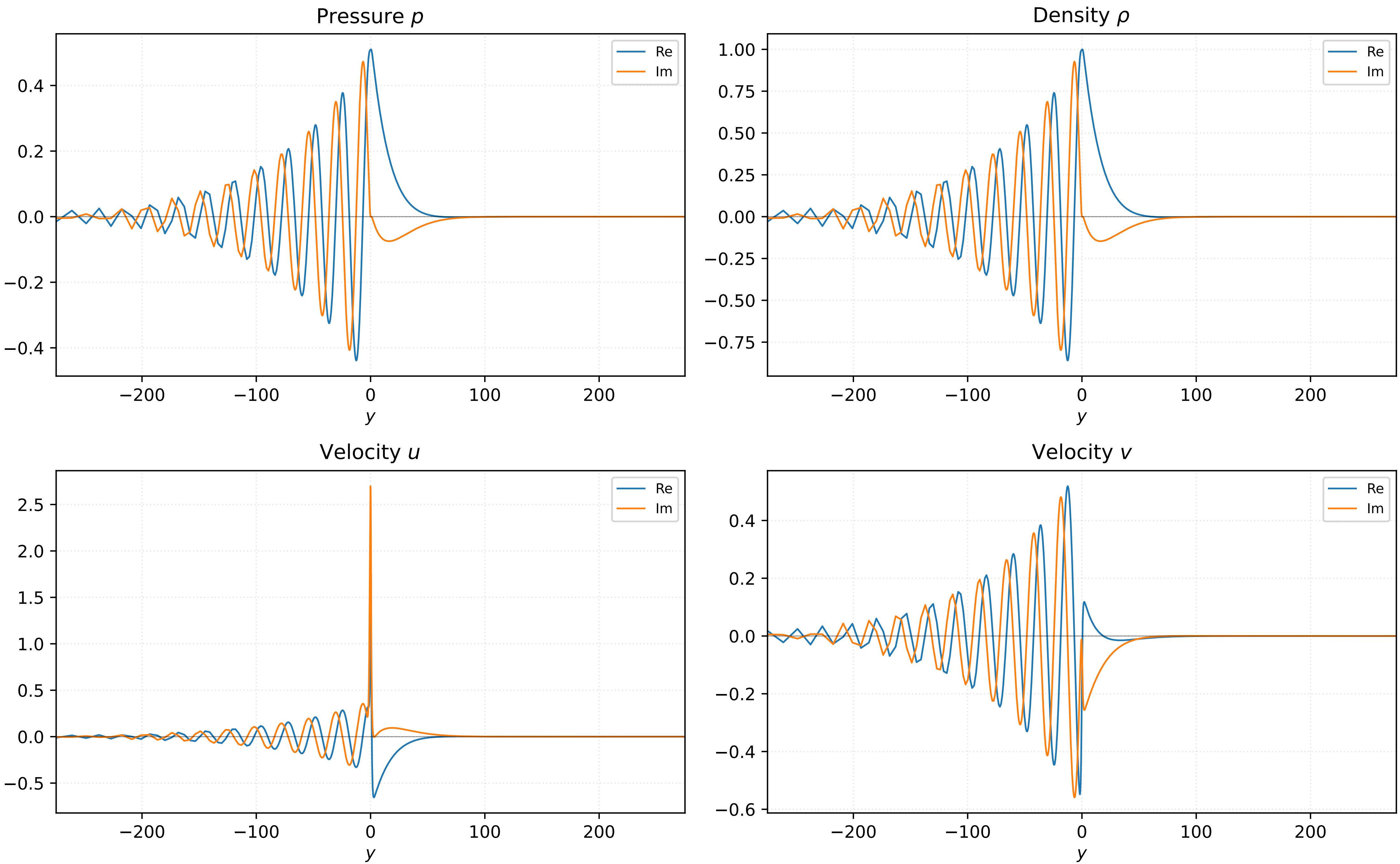

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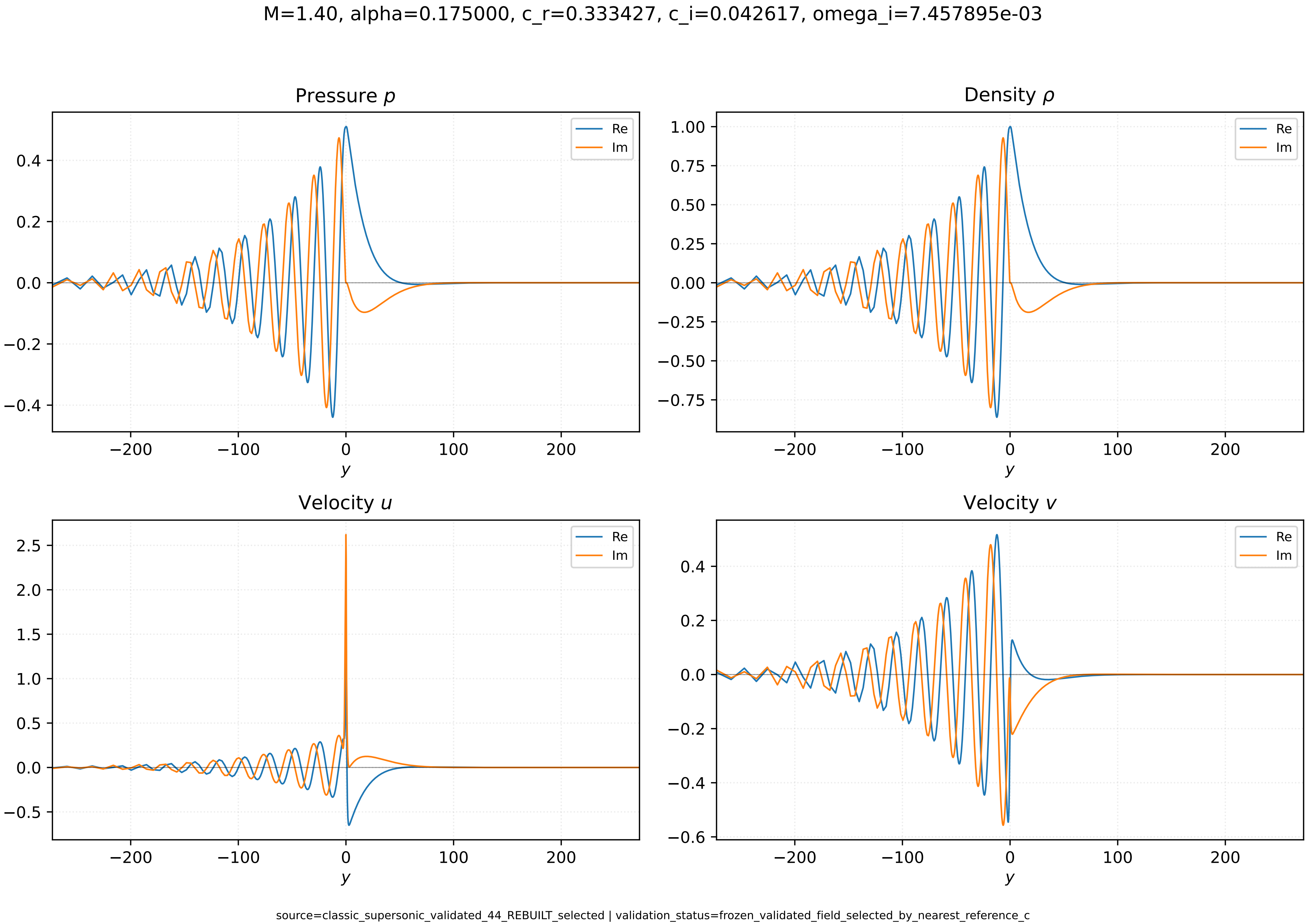
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Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
y
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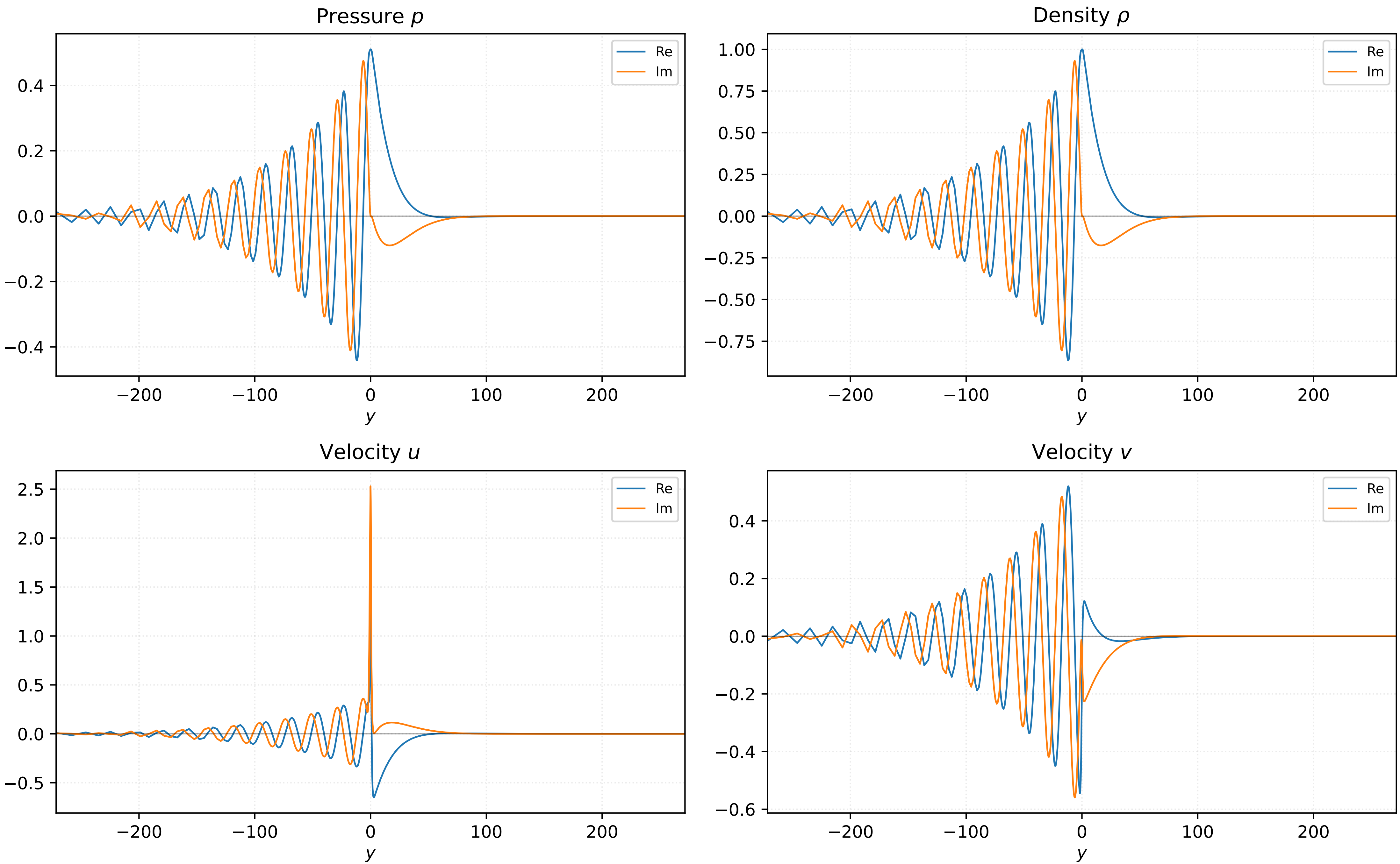

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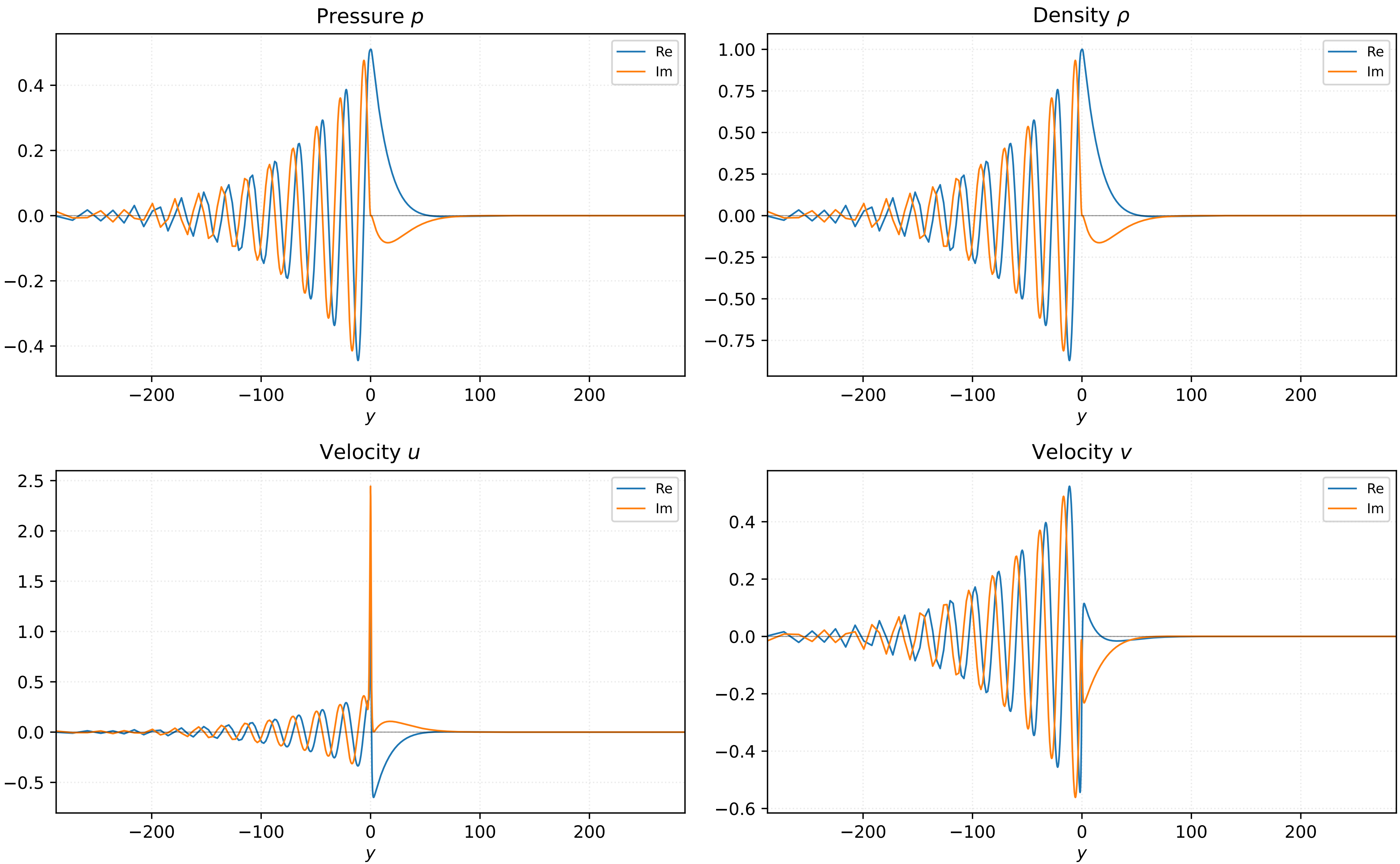

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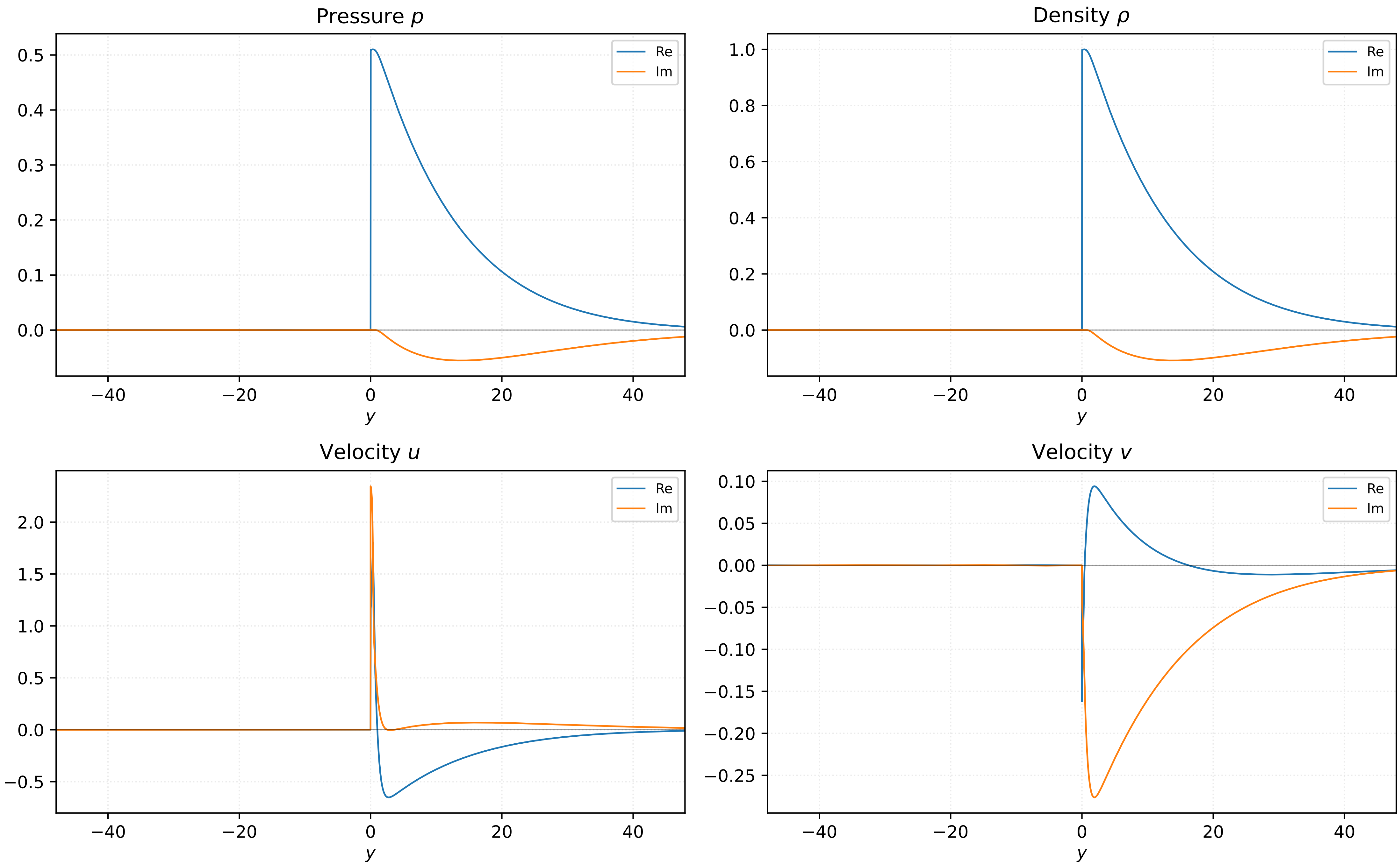

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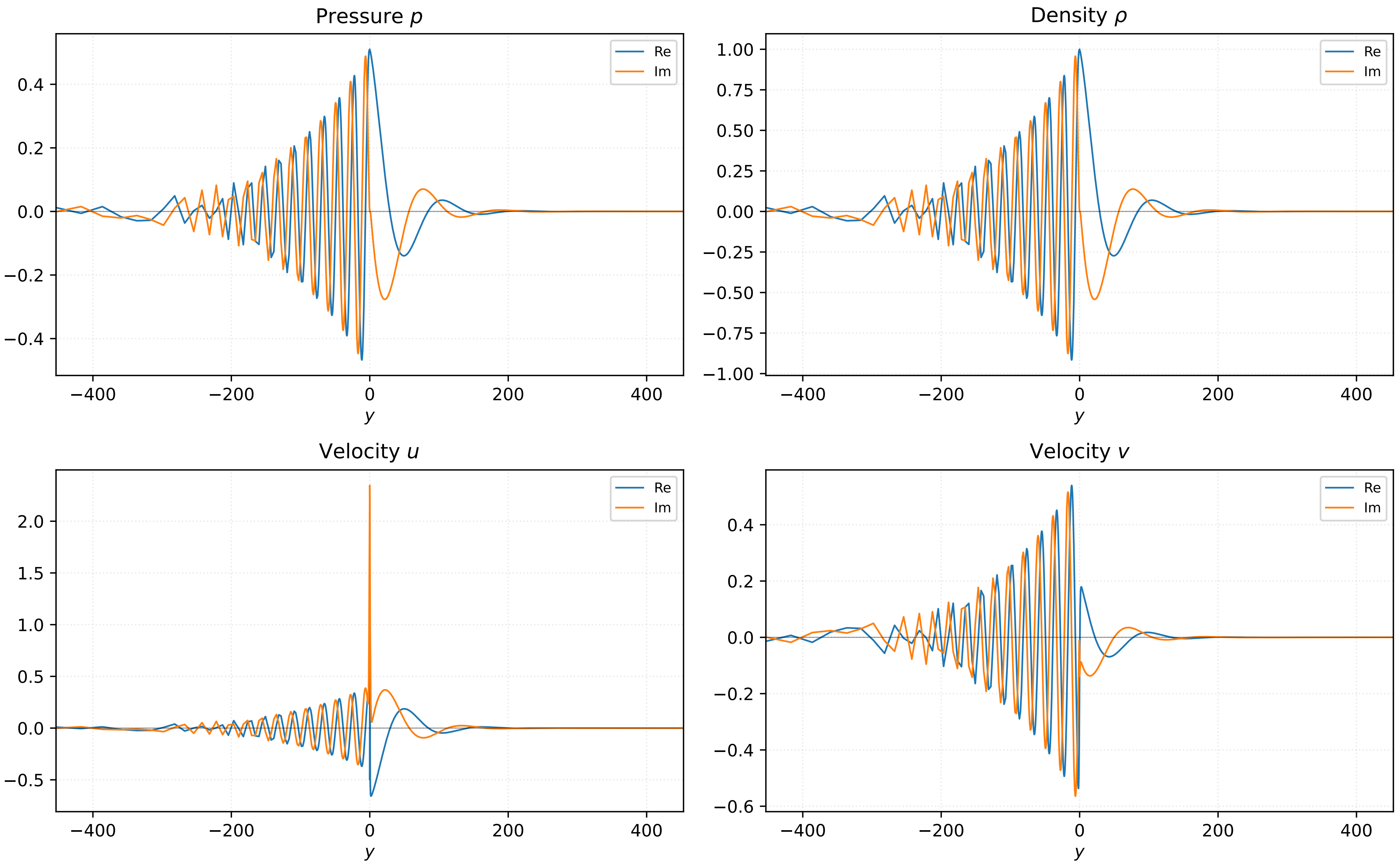

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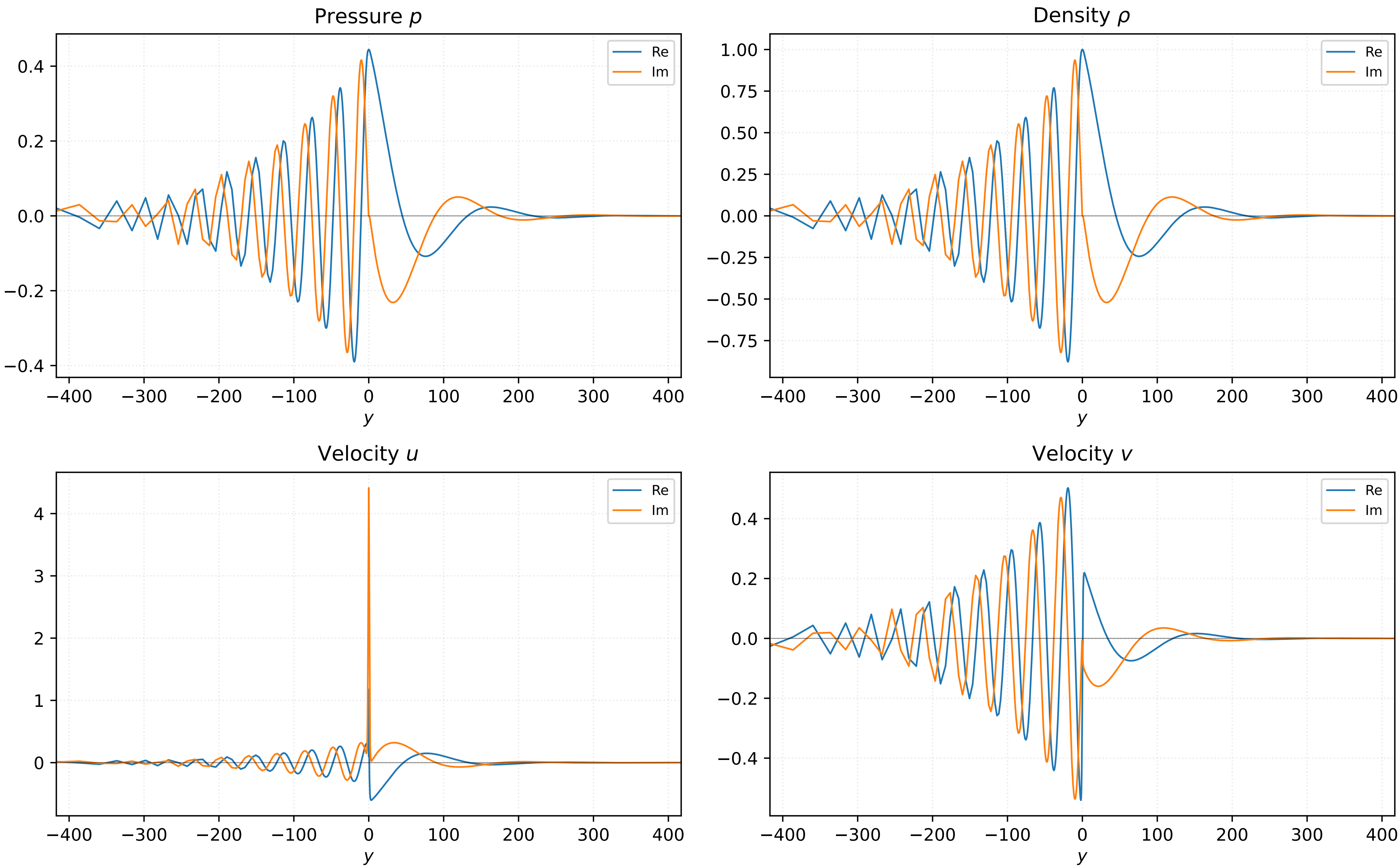

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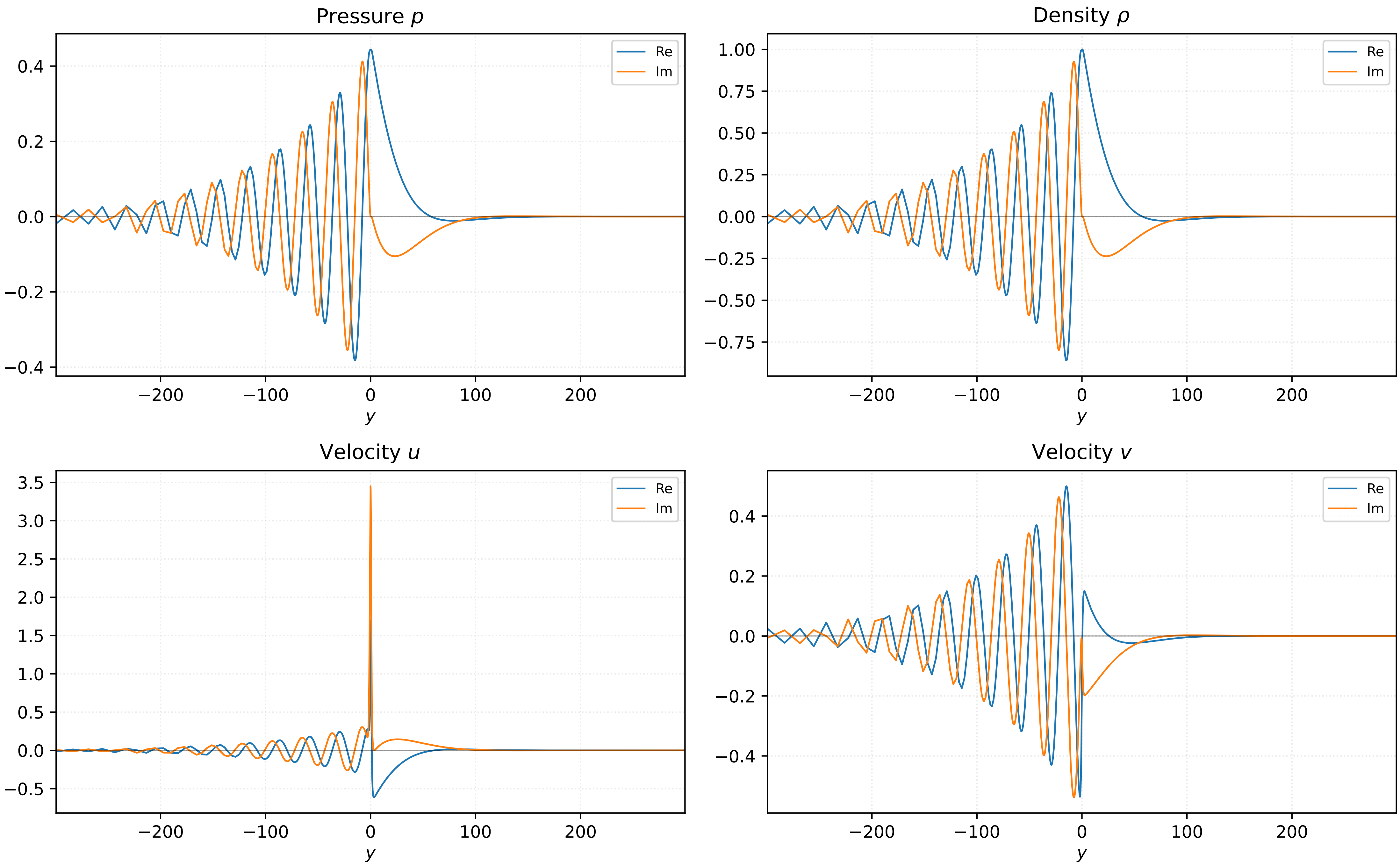

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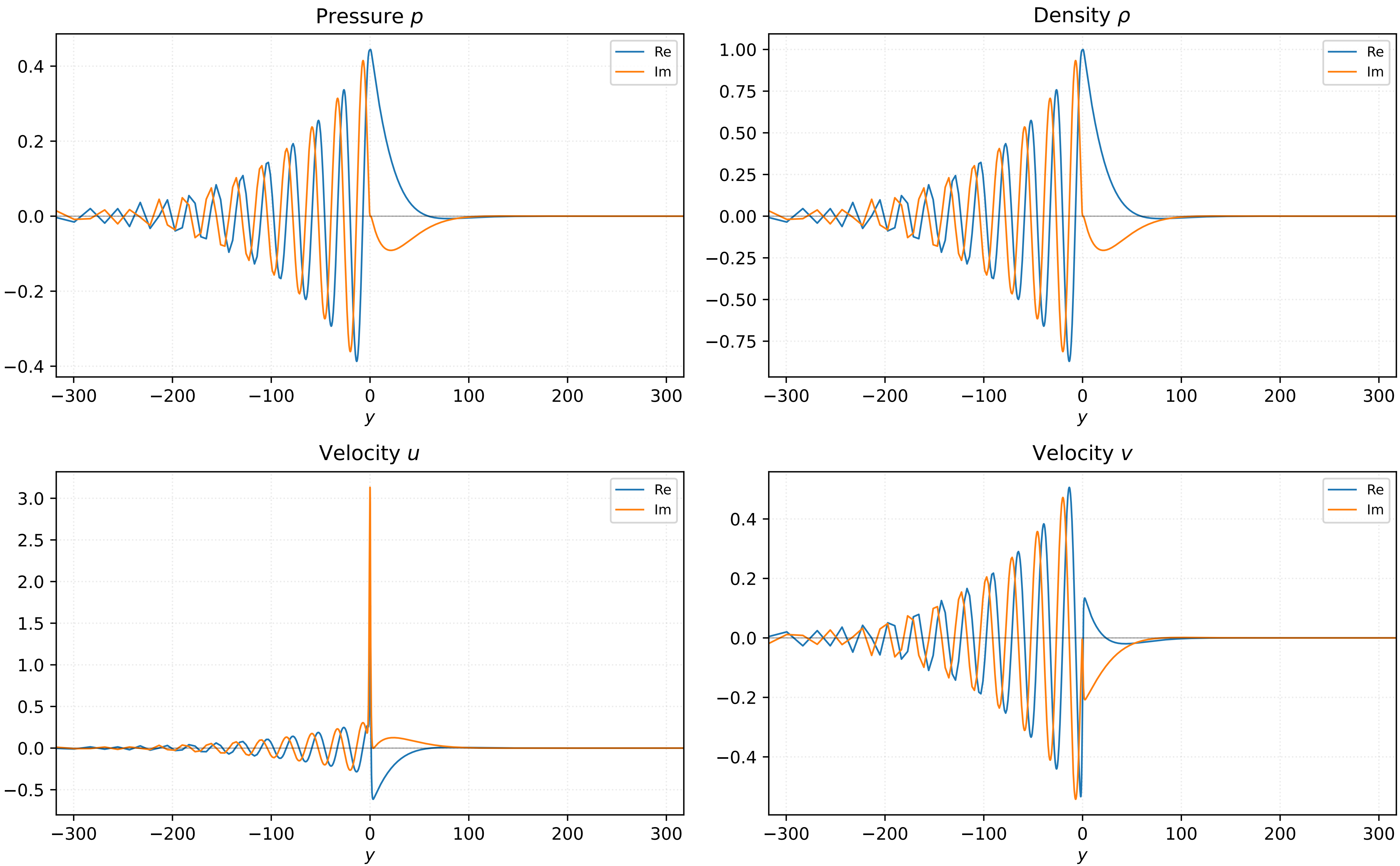

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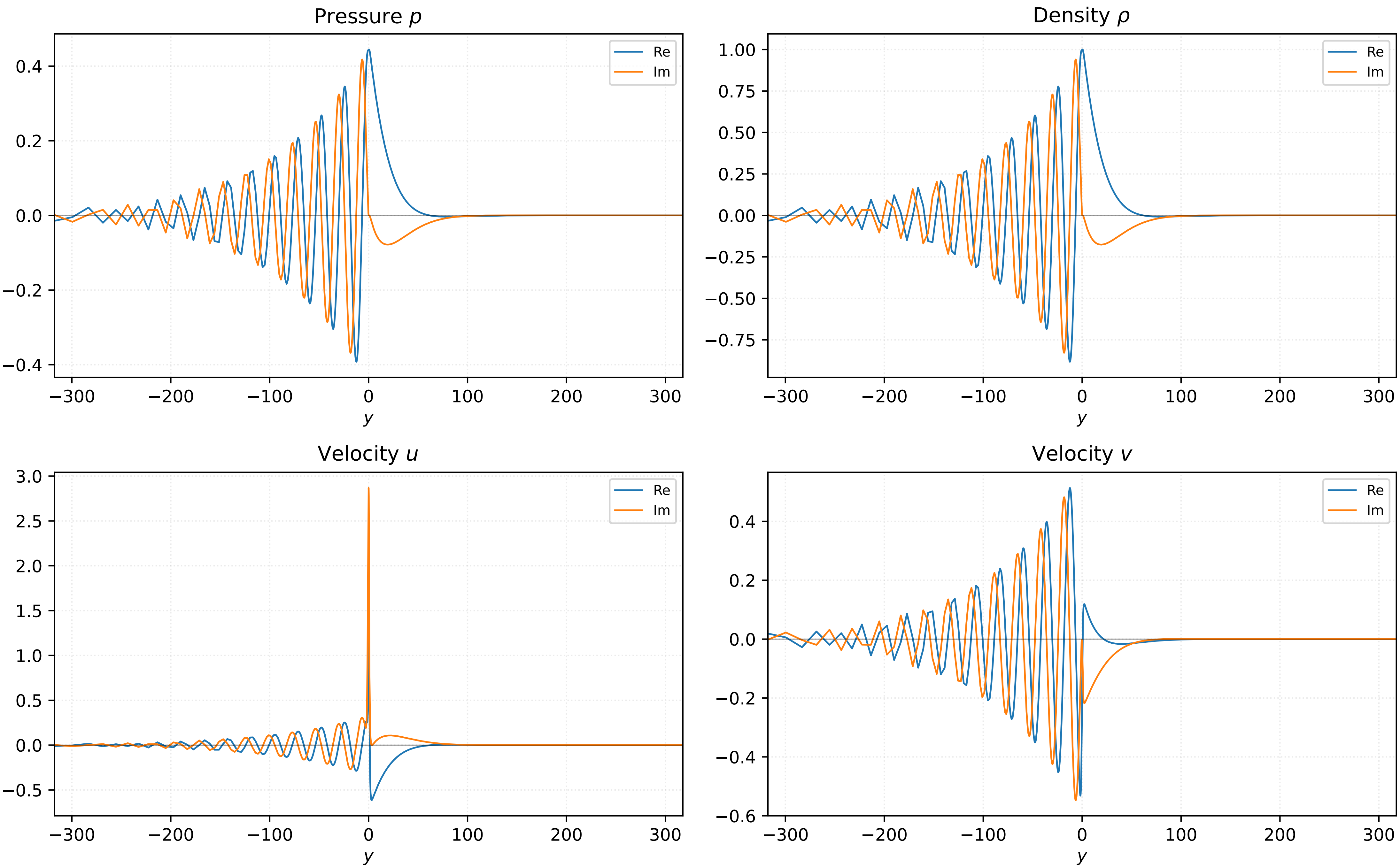

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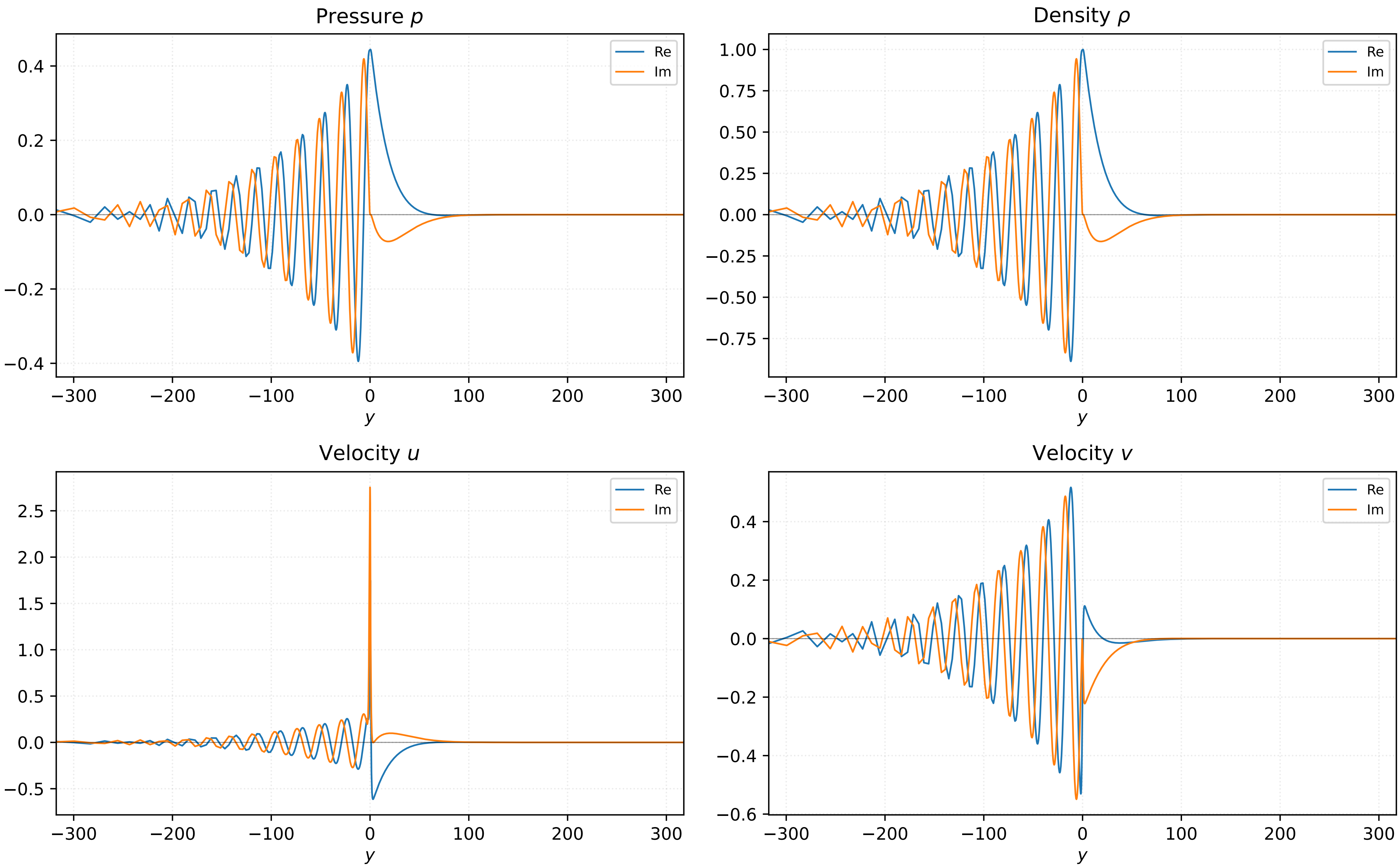

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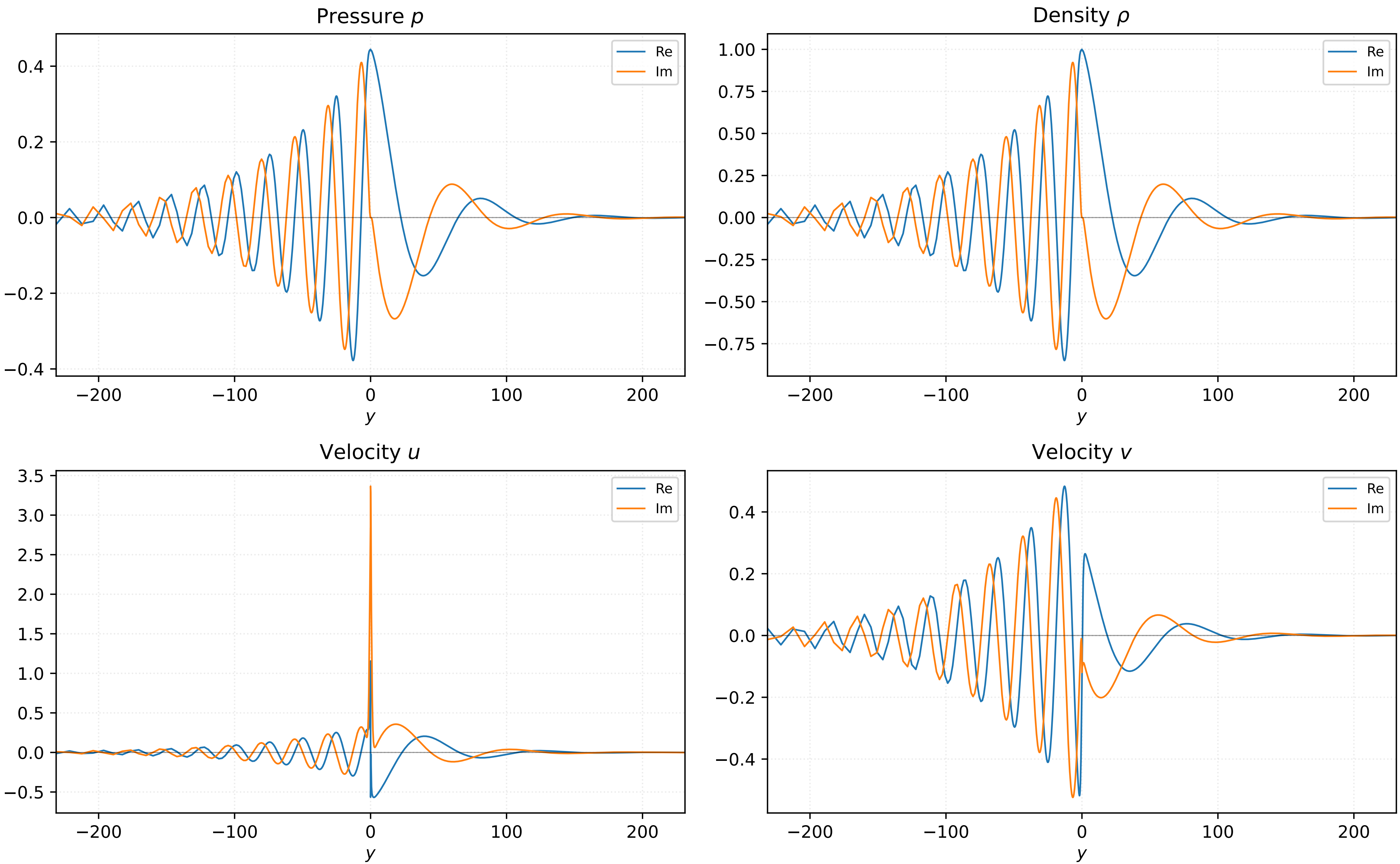

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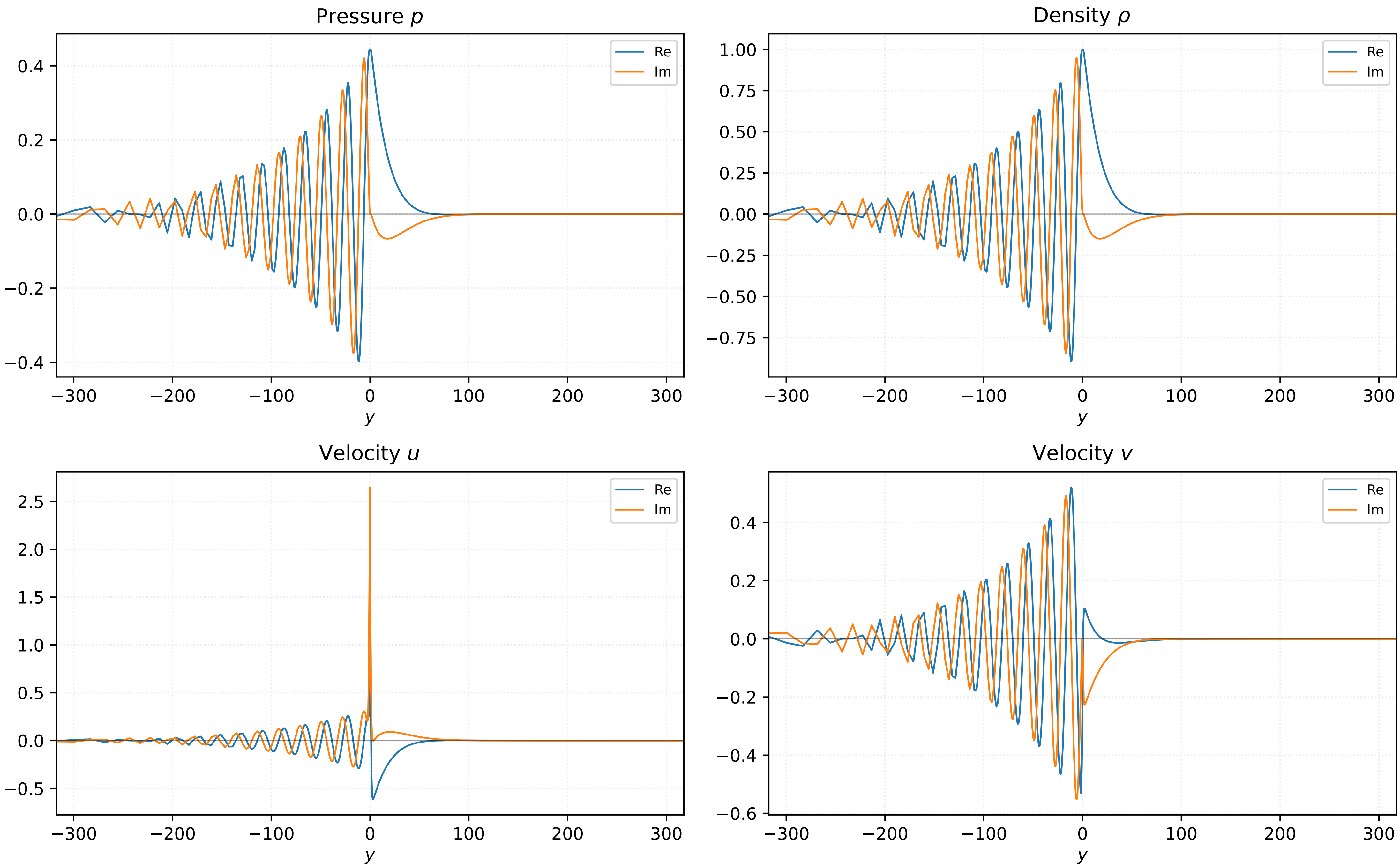

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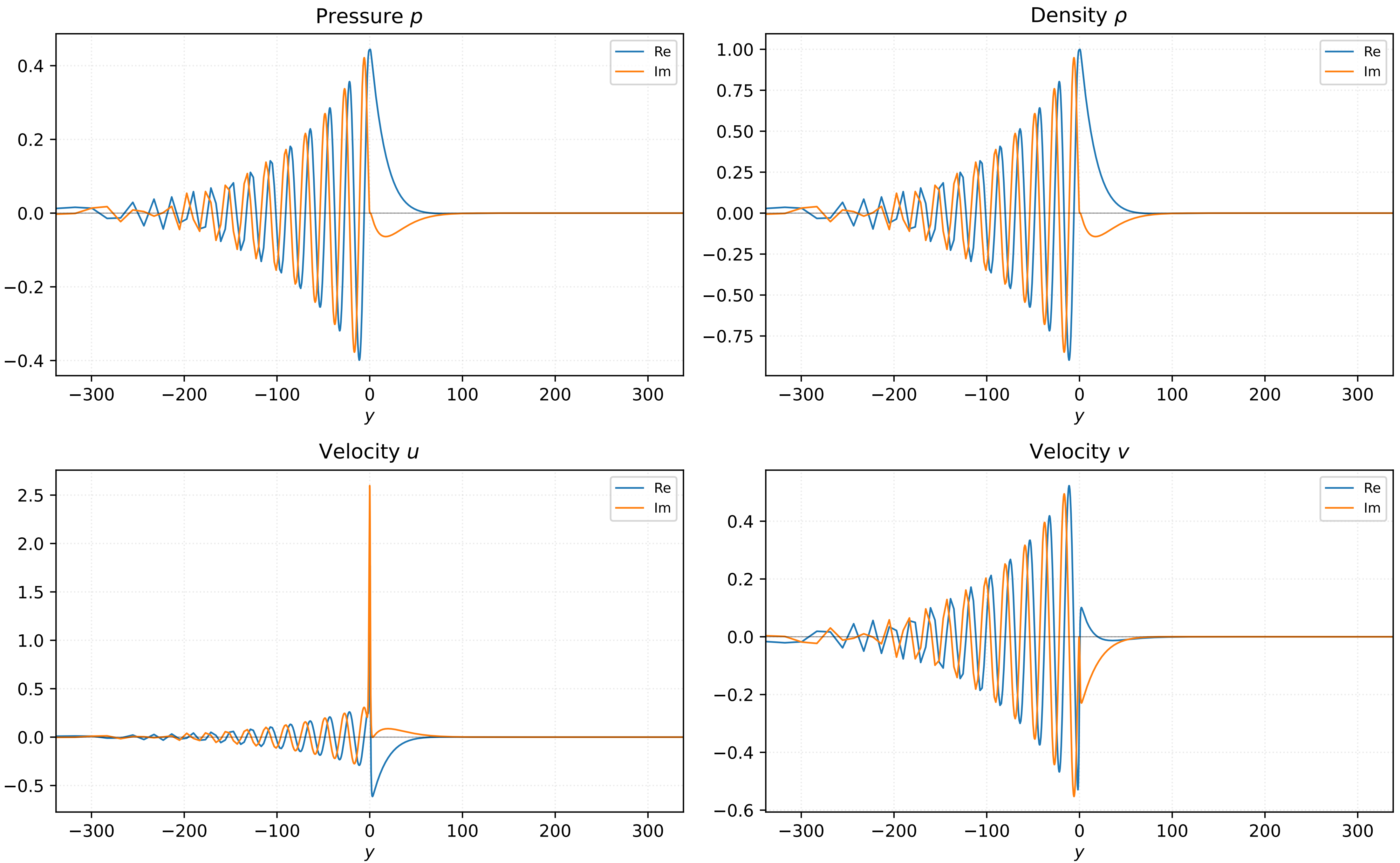

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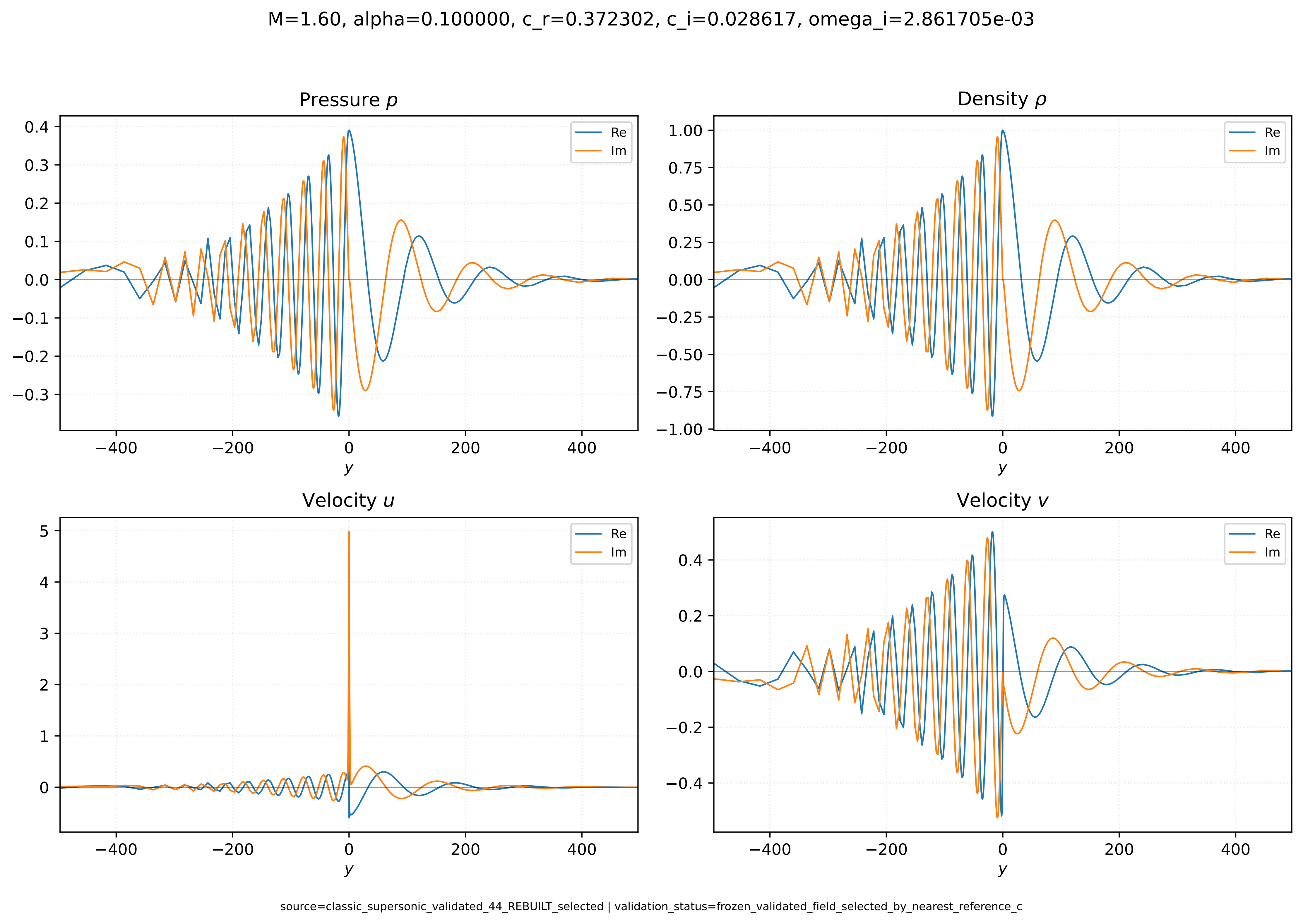
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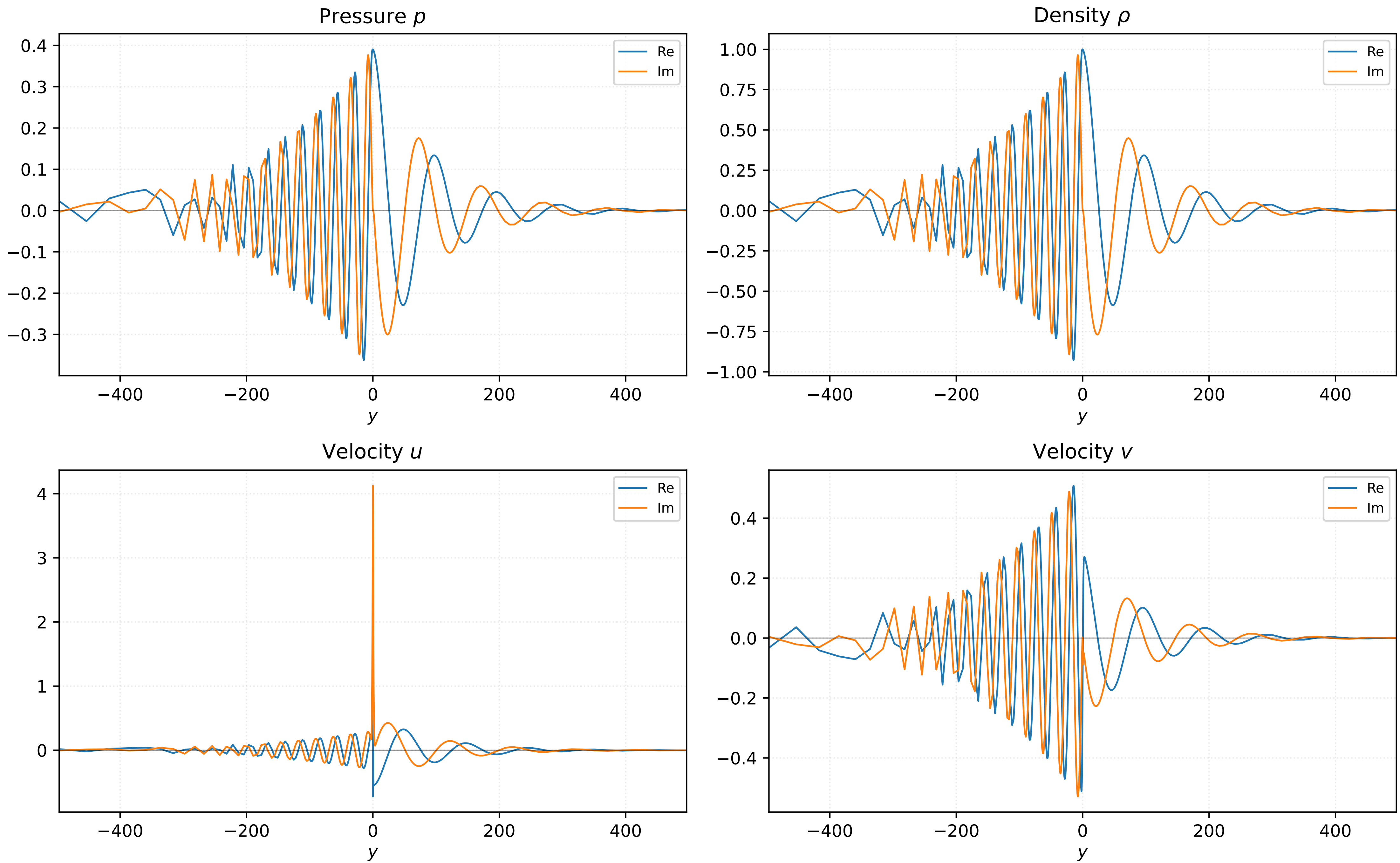
Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
y
source=classic_supersonic_validated_44_REBUILT_selected | validation_status=frozen_validated_field_selected_by_nearest_reference_c

M=1.60, alpha=0.125000, c_r=0.387564, c_i=0.025110, omega_i=3.138734e-03



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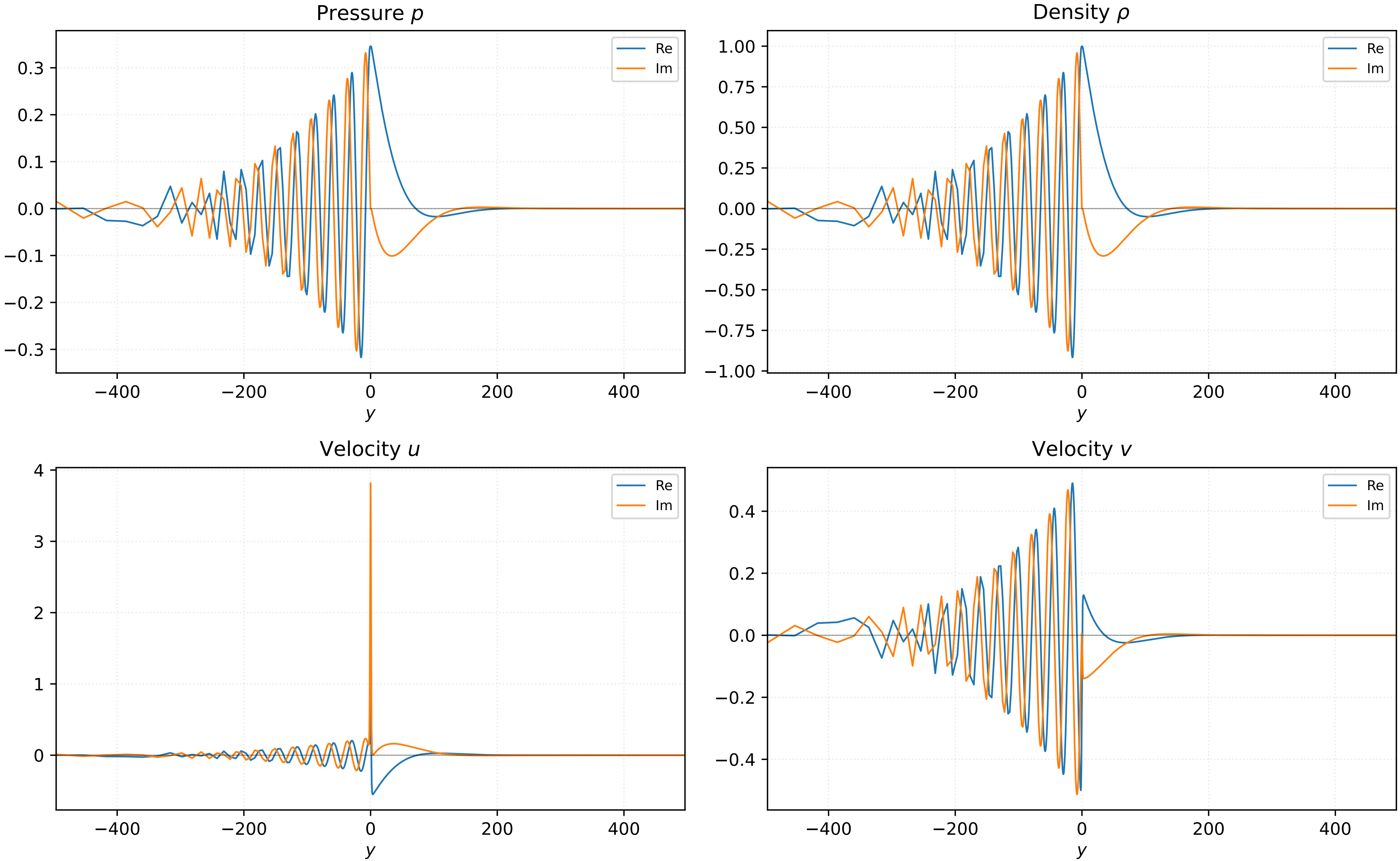
M=1.70, alpha=0.100000, c_r=0.408672, c_i=0.025583, omega_i=2.558329e-03



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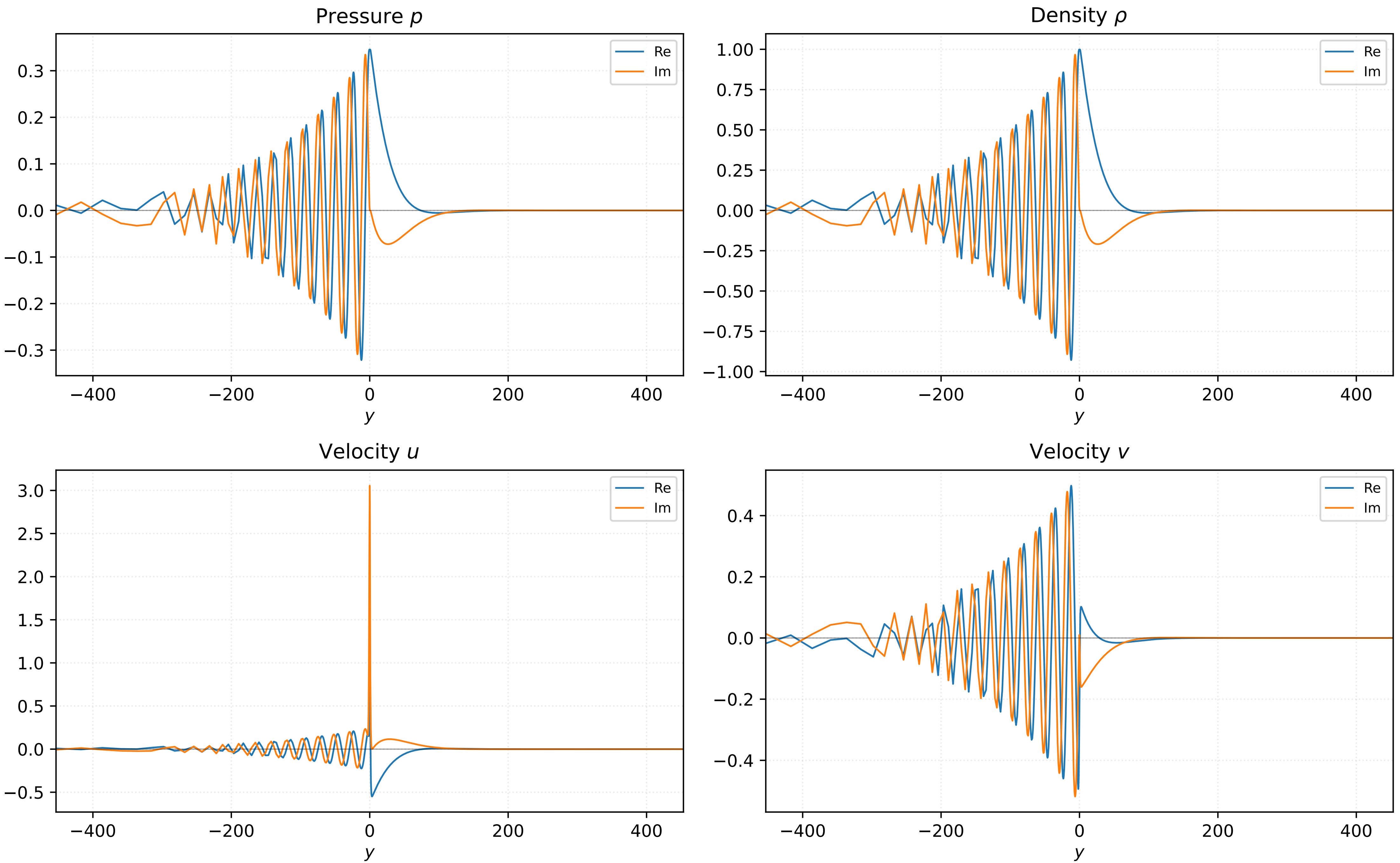

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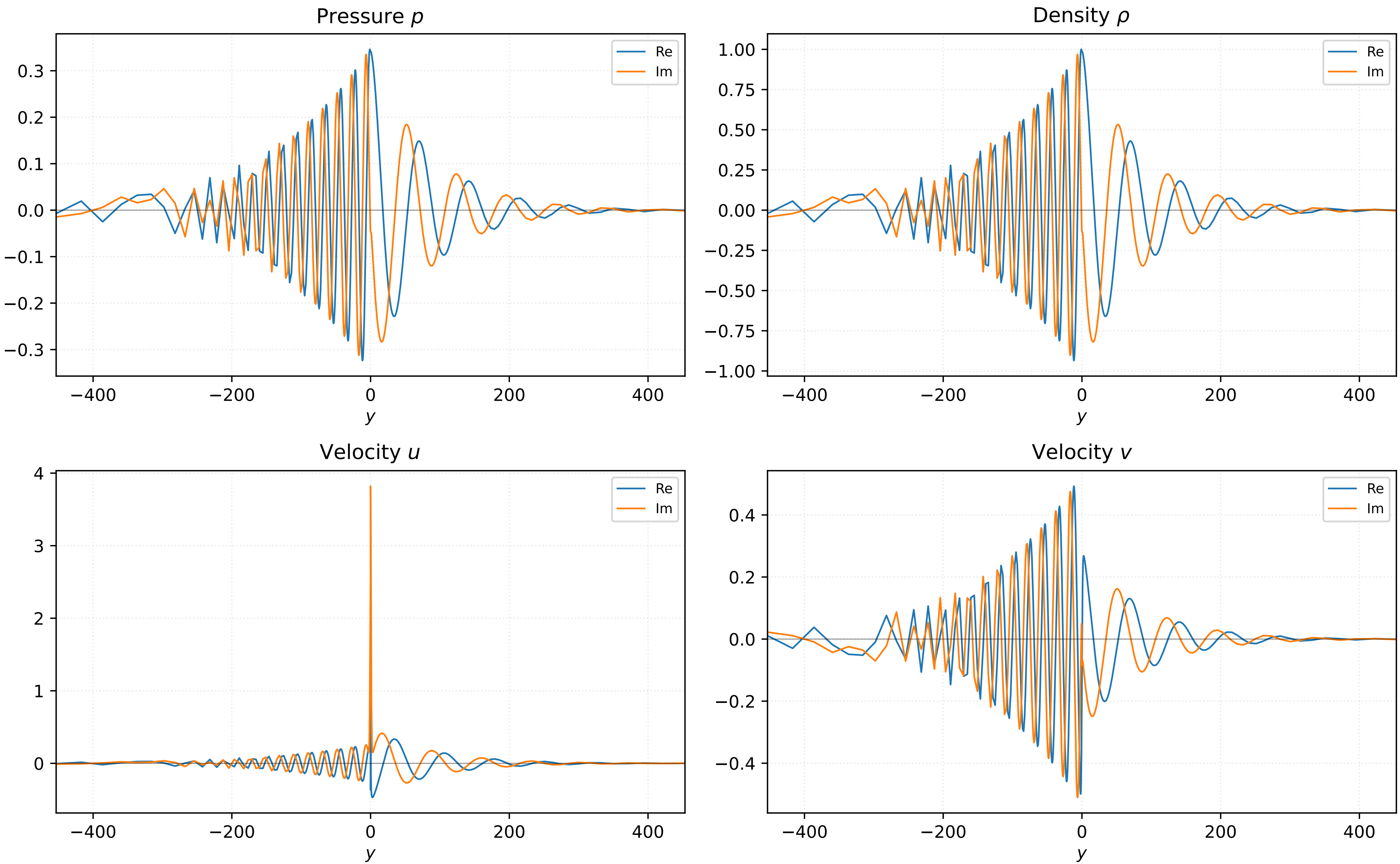

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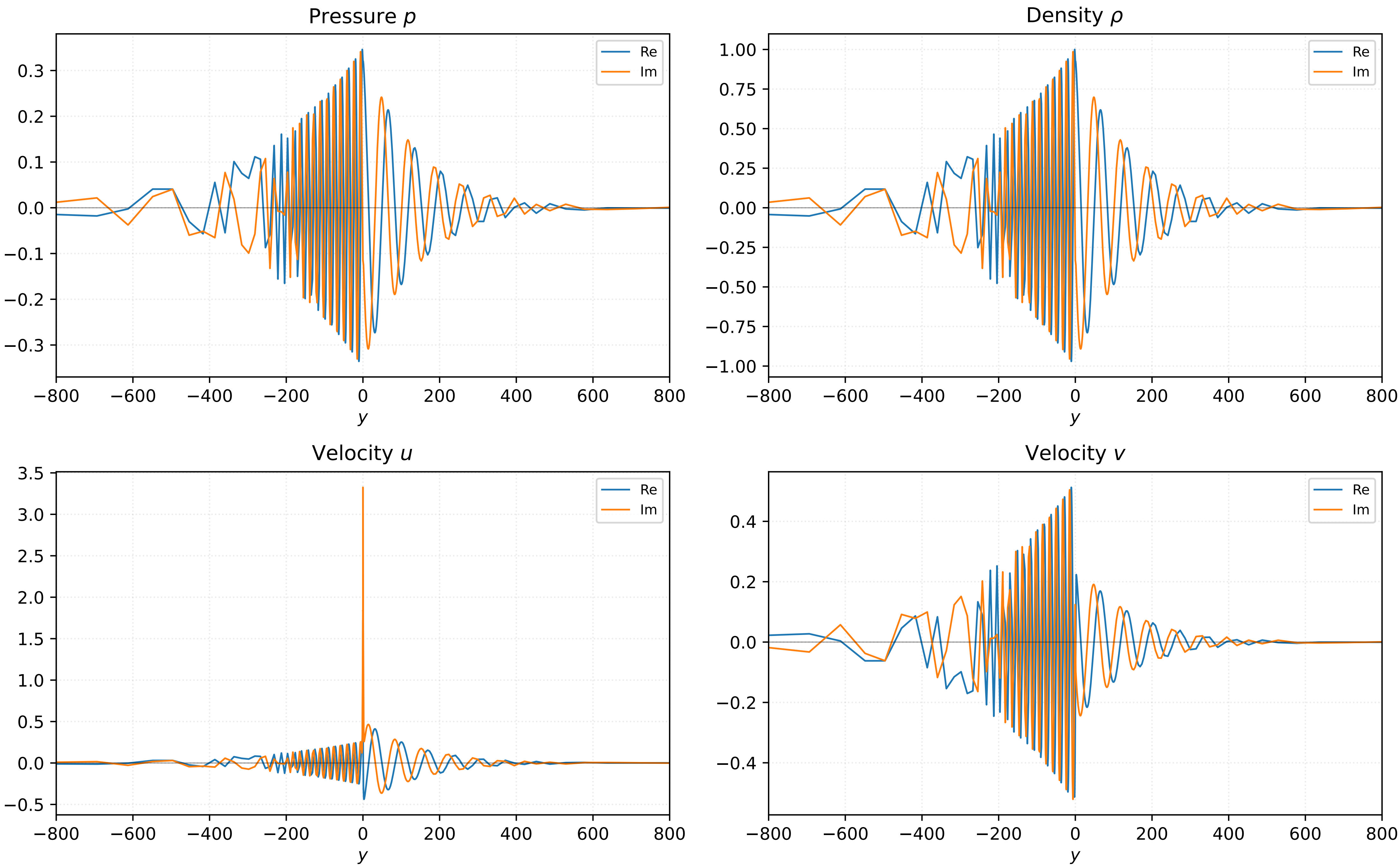
M=1.70, alpha=0.175000, c_r=0.440380, c_i=0.015395, omega_i=2.694203e-03



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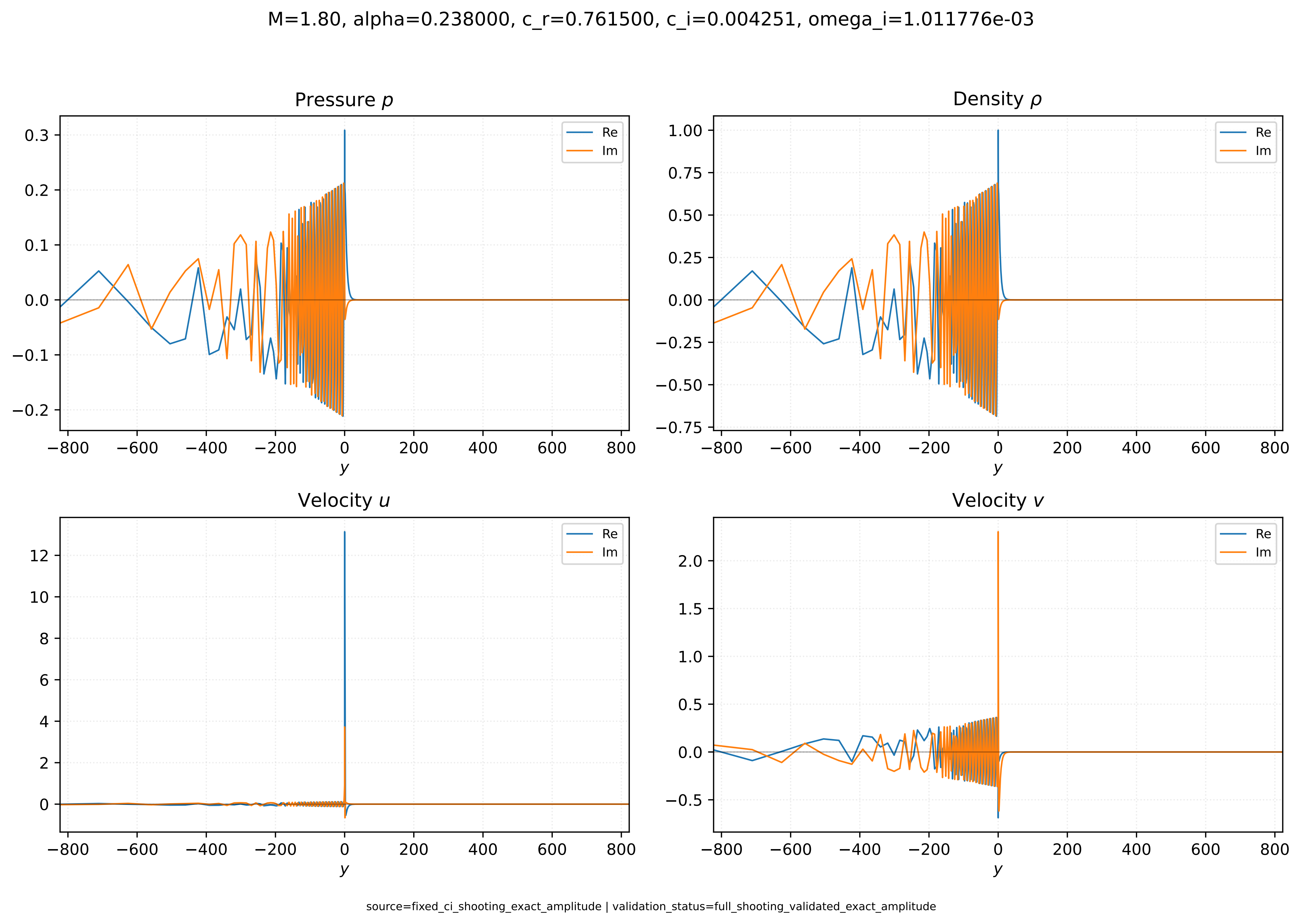
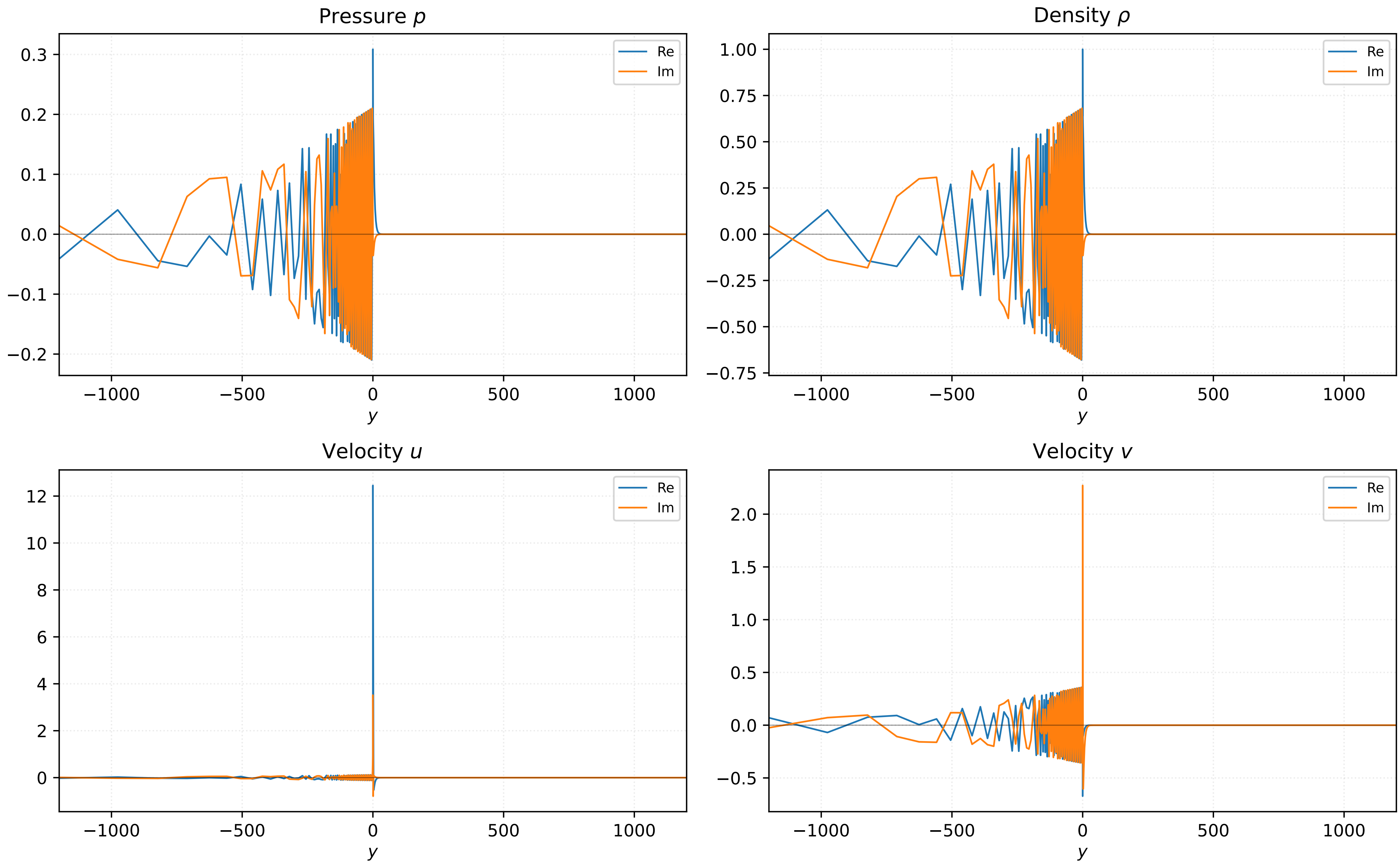


M=1.80, alpha=0.238000, c_r=0.761500, c_i=0.004251, omega_i=1.011776e-03
Pressure p
Density ρ
Velocity u
Velocity v
Re
Im
y
source=fixed_ci_shooting_exact_amplitude | validation_status=full_shooting_validated_exact_amplitude

M=1.80, alpha=0.248000, c_r=0.761553, c_i=0.002802, omega_i=6.950027e-04



source=fixed_ci_shooting_exact_amplitude | validation_status=full_shooting_validated_exact_amplitude

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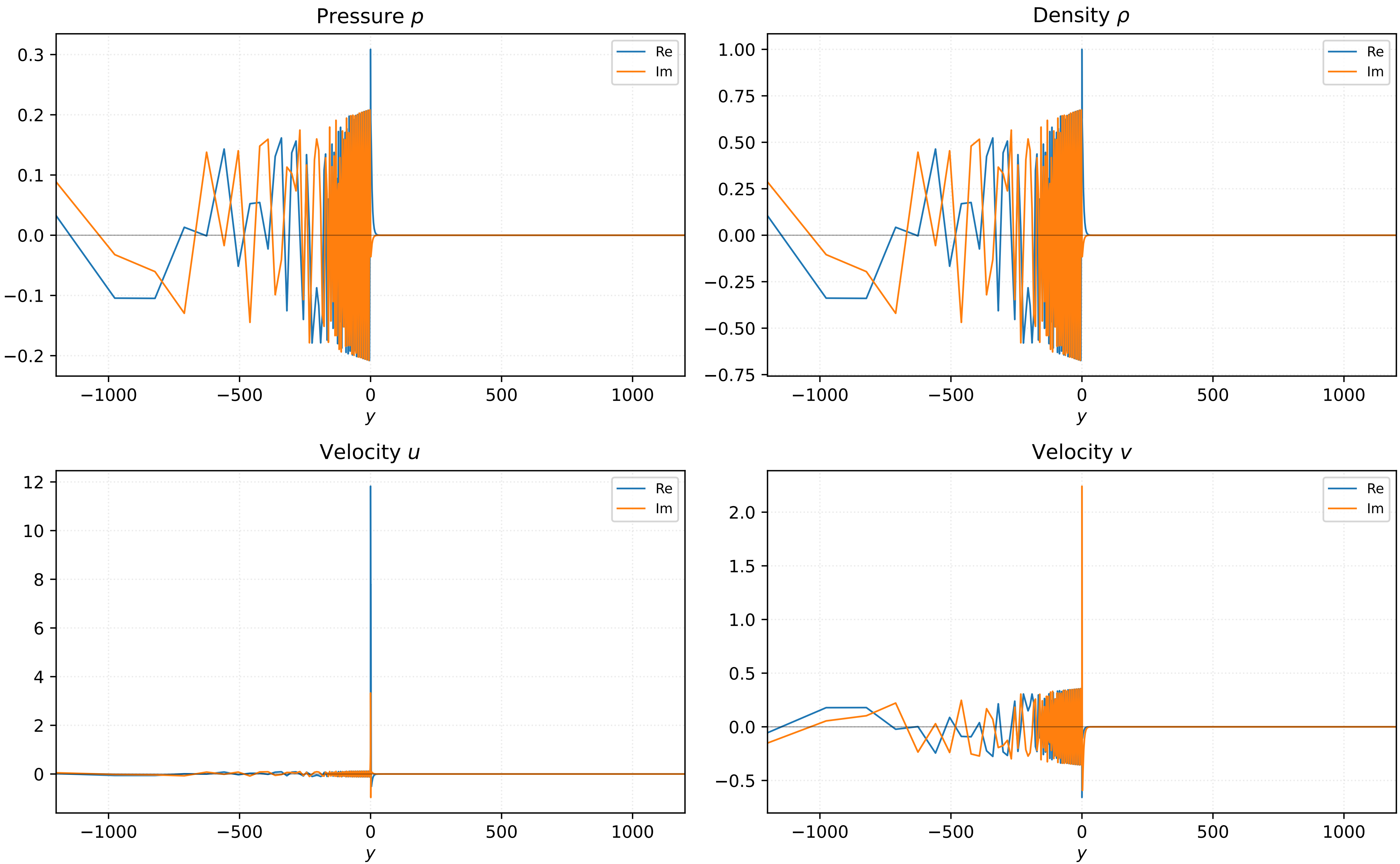
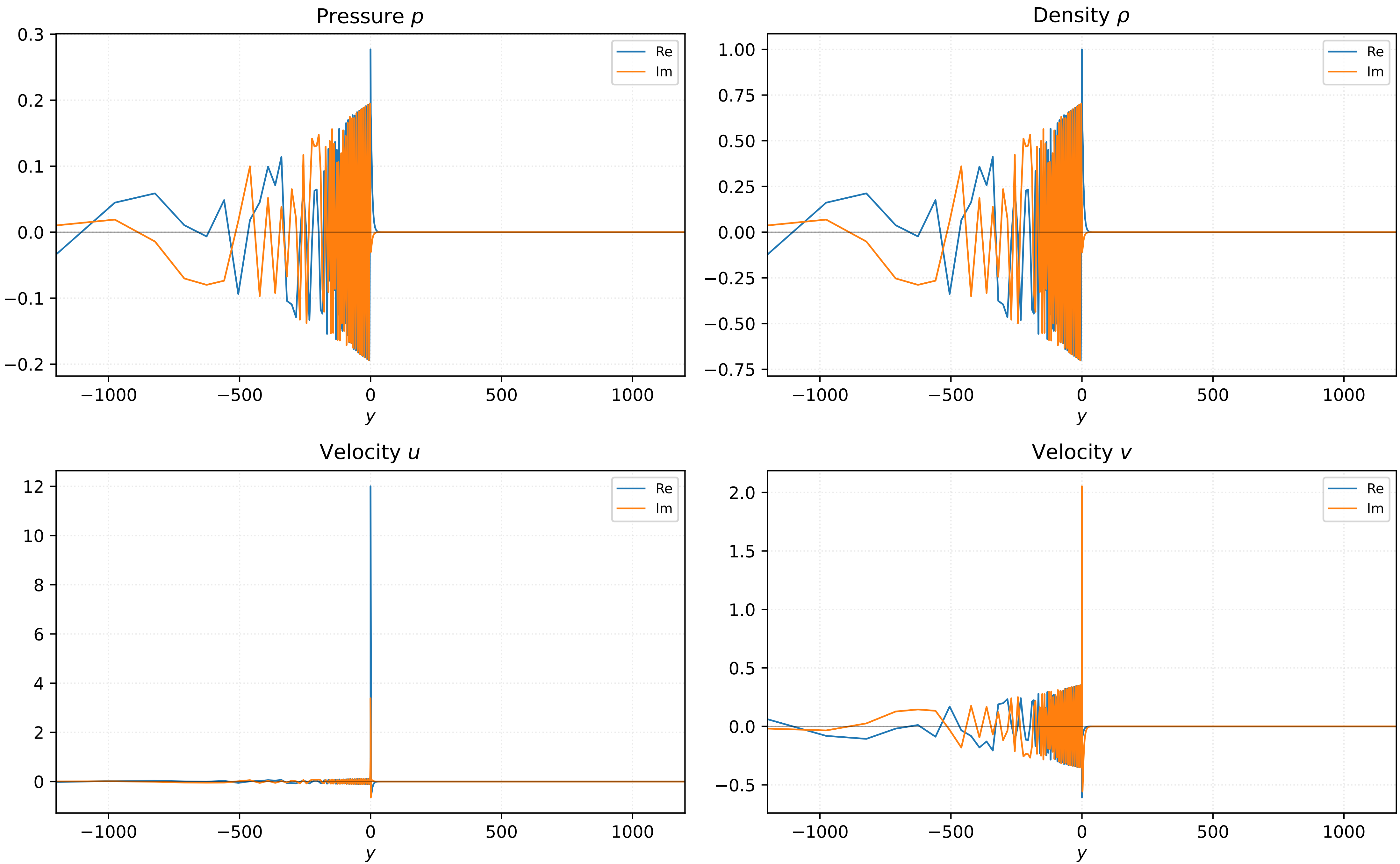


source=fixed_ci_shooting_exact_amplitude | validation_status=full_shooting_validated_exact_amplitude

M=1.90, alpha=0.232000, c_r=0.761543, c_i=0.003092, omega_i=7.174413e-04



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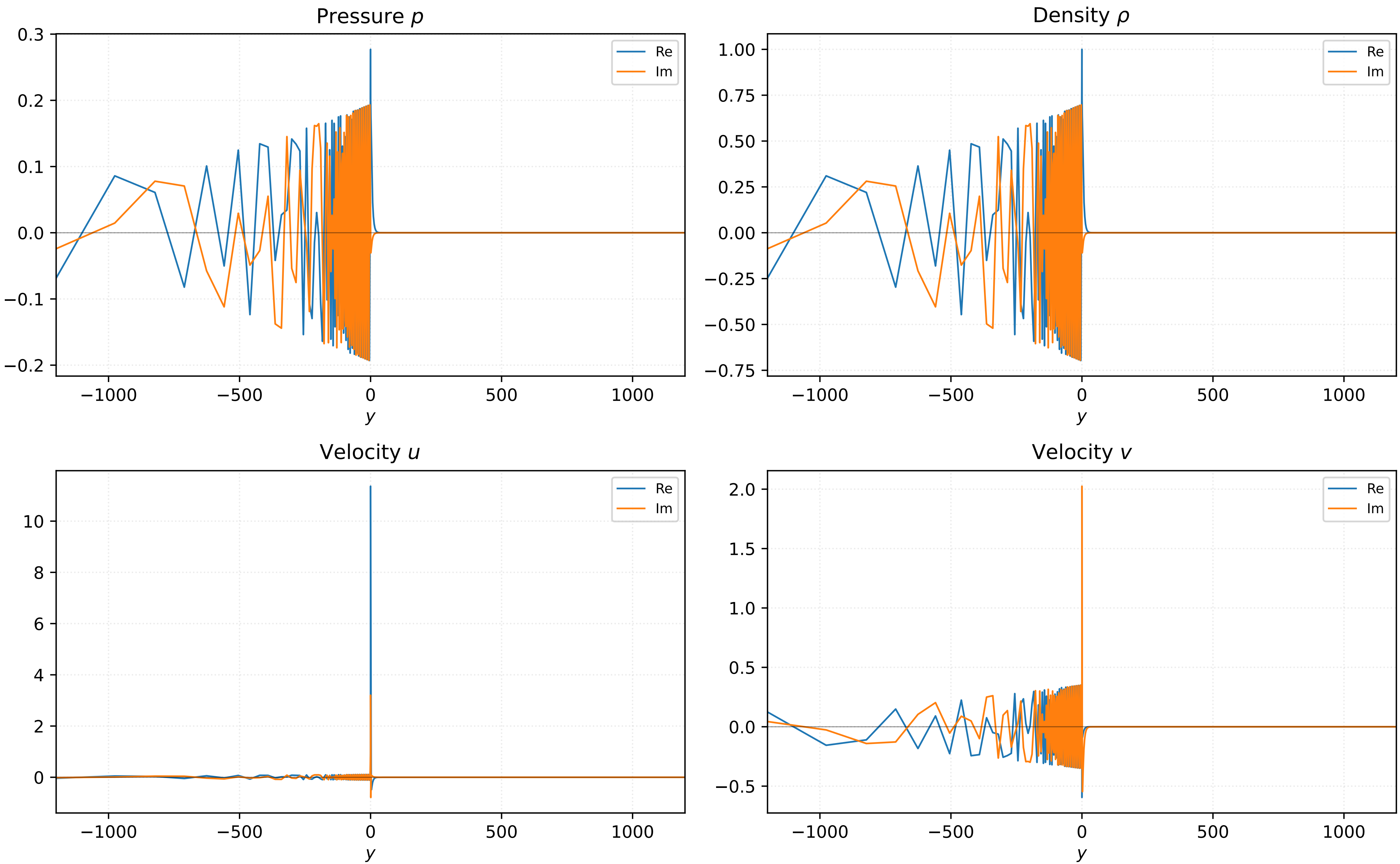
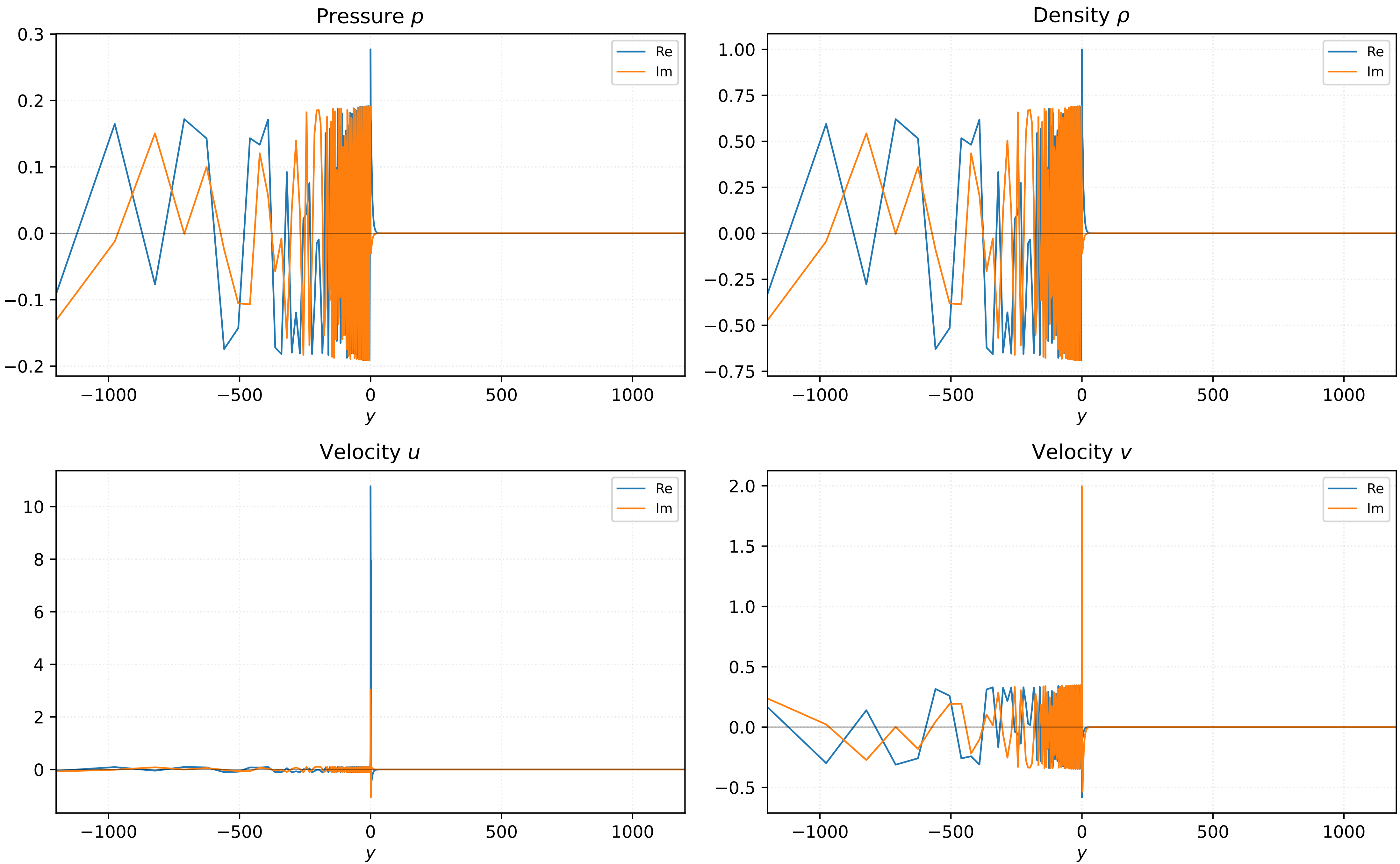


source=fixed_ci_shooting_exact_amplitude | validation_status=full_shooting_validated_exact_amplitude

M=1.90, alpha=0.252000, c_r=0.761594, c_i=0.000304, omega_i=7.664211e-05



source=fixed_ci_shooting_exact_amplitude | validation_status=full_shooting_validated_exact_amplitude